ISC Class XII Mathematics

Chapter 10
Vector Algebra

51 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

Compiled Notes – Formula Sheet + Type-wise Solved ISC Board Solutions

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Formula Sheet

Section I – Basics

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Magnitude of $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

Section I – Basics

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

Unit vector in the direction of $\vec{a}$

Section I – Basics

$$\vec{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

Vector joining $P(x_1, y_1, z_1)$ to $Q(x_2, y_2, z_2)$

Section I – Basics

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

Relation between direction cosines $l = \cos\alpha,\ m = \cos\beta,\ n = \cos\gamma$

Section II – Position Vectors

$$\vec{M} = \frac{\vec{a} + \vec{b}}{2}$$

Position vector of midpoint of segment joining points with p.v. $\vec{a}, \vec{b}$

Section II – Position Vectors

$$\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

Centroid of a triangle with vertex p.v. $\vec{a}, \vec{b}, \vec{c}$

Section II – Position Vectors

$$\vec{R} = \frac{m\vec{b} + n\vec{a}}{m + n}$$

Internal division of $\vec{a}, \vec{b}$ in ratio $m : n$

Section II – Position Vectors

$$\vec{R} = \frac{m\vec{b} - n\vec{a}}{m - n}$$

External division of $\vec{a}, \vec{b}$ in ratio $m : n$

Section III – Collinearity

$$\vec{AB} = \lambda\,\vec{AC}$$

Points A, B, C collinear $\iff$ $\vec{AB}$ and $\vec{AC}$ are parallel (scalar multiples)

Section IV – Dot Product

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

Definition; θ is the angle between $\vec{a}$ and $\vec{b}$

Section IV – Dot Product

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Dot product in component form

Section IV – Dot Product

$$\text{Projection of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

Scalar projection formula

Section IV – Dot Product

$$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$$

Perpendicularity test using the dot product

Section V – Cross Product

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta\,\hat{n}$$

Definition; $\hat{n}$ is the unit normal (right-hand rule)

Section V – Cross Product

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Cross product in component (determinant) form

Section V – Cross Product

$$\text{Area of parallelogram} = |\vec{a} \times \vec{b}|$$

With $\vec{a}, \vec{b}$ as adjacent sides

Section V – Cross Product

$$\text{Area of triangle} = \frac{1}{2}|\vec{a} \times \vec{b}|$$

With $\vec{a}, \vec{b}$ as two sides from the same vertex

Section V – Cross Product

$$\text{Area of parallelogram} = \frac{1}{2}|\vec{d_1} \times \vec{d_2}|$$

With $\vec{d}_1, \vec{d}_2$ as the diagonals

Section V – Cross Product

$$\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0}$$

Parallel-vector test using the cross product

I. Vector Fundamentals

Type 1

Classifying physical quantities as scalars (magnitude only) or vectors (magnitude *and* direction). Recognise the type by asking: does the quantity need a direction to be fully specified?

Source: Exercise 10.1, Q1

Given: Classify as scalar or vector: (i) 5 seconds (ii) 20 m/sec^2 (iii) 1000 cm^3 (iv) 2 metres north-west (v) 40 watt (vi) 2 km

Solution:

Check each quantity for a direction component.
(i) Time has magnitude only $\Rightarrow$ scalar.
(ii) Acceleration has both size and direction $\Rightarrow$ vector.
(iii) Volume has magnitude only $\Rightarrow$ scalar.
(iv) "2 metres north-west" states both a distance and a direction $\Rightarrow$ vector (this is displacement).
(v) Power (watt) has magnitude only $\Rightarrow$ scalar.
(vi) "2 km" alone (no direction stated) is just distance $\Rightarrow$ scalar.

Final Answer

(i) Scalar (ii) Vector (iii) Scalar (iv) Vector (v) Scalar (vi) Scalar

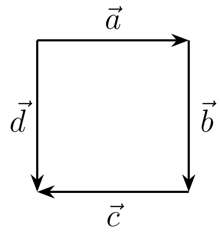
Common Mistake: Students often mark (iv) as scalar because it "looks like" a distance – the stated direction (north-west) is exactly what makes it a vector (displacement, not distance).

Quick Recall: If a direction is stated or implied, it's a vector; if only a size is given, it's a scalar.

Type 2

Reading vector relationships (equal, co-initial, collinear-but-not-equal) directly off a given figure. Equal vectors need same magnitude *and* direction; co-initial vectors share a starting point; collinear vectors lie along the same or parallel lines.

Source: Exercise 10.1, Q2



Square with $\vec{a}$ (top, pointing right), $\vec{b}$ (right side, pointing down), $\vec{c}$ (bottom, pointing left), $\vec{d}$ (left side, pointing down)

Given: A square with sides labelled $\vec{a}$ (top), $\vec{b}$ (right), $\vec{c}$ (bottom), $\vec{d}$ (left), directions as drawn.

Solution:

Equal vectors need identical length AND identical direction, not just parallel sides.
$\vec{b}$ points downward on the right side; $\vec{d}$ points downward on the left side – same length, same direction $\Rightarrow$ $\vec{b}$ and $\vec{d}$ are **equal**.
Co-initial vectors share the same starting (tail) point.
$\vec{a}$ starts at the top-left corner and $\vec{d}$ also starts at the top-left corner $\Rightarrow$ $\vec{a}$ and $\vec{d}$ are **co-initial**.
Collinear vectors lie on the same or parallel lines but need not be equal.
$\vec{a}$ (top, left to right) and $\vec{c}$ (bottom, right to left) lie on parallel lines but point in opposite senses and are traversed oppositely $\Rightarrow$ $\vec{a}$ and $\vec{c}$ are **collinear but not equal**.

Final Answer

Equal: $\vec{b}, \vec{d}$ Co-initial: $\vec{a}, \vec{d}$ Collinear but not equal: $\vec{a}, \vec{c}$

Common Mistake: Calling opposite sides of a square "equal vectors" just because they have the same length – direction must also match exactly.

Quick Recall: Equal = same length & same direction; co-initial = same tail; collinear = same/parallel line, direction can differ.

Type 3

True/False conceptual statements testing the precise definitions of equal, collinear, proper, and zero vectors, and the algebra of scalar multiplication with vectors. Recognise this type from a list of short conceptual claims with no computation required.

Source: Exercise 10.1, Q3, Q4

Given: 14 statements (Q3) and 4 statements (Q4) about properties of vectors $\vec{a}, \vec{b}$ – see textbook for the full list.

Solution:

Go through each statement testing it against the strict definitions – a single counterexample is enough to make a statement False.
$\vec{a}$ and $-\vec{a}$ point in opposite directions but lie on the same line $\Rightarrow$ they are collinear: **True**.
Two collinear vectors can have completely different magnitudes (only their lines need to match) $\Rightarrow$ **False**.
Two vectors with the same magnitude can point in totally different directions, so they need not be collinear $\Rightarrow$ **False**.
Two collinear vectors with the same magnitude could still point in opposite directions, so they need not be equal $\Rightarrow$ **False**.
The zero vector has no fixed direction, so "unique" here refers to it being the unique additive identity $\Rightarrow$ **False** (direction is not unique/defined).
$3\vec{a}$ and $-3\vec{a}$ have the same magnitude $3|\vec{a}|$ but opposite direction, so their moduli (magnitudes) are equal $\Rightarrow$ **True**.
$\vec{a} + (-\vec{a}) = \vec{0}$, which is a zero vector, not a "proper" (non-zero) vector $\Rightarrow$ **False**.
Equal and co-initial vectors share both magnitude, direction, AND starting point – their endpoints must then coincide too $\Rightarrow$ **True**.
Distributivity of scalar addition/multiplication over vectors always holds $\Rightarrow$ **True** for (ix), **False** for (x) since $(mn)(\vec{a} + \vec{b}) \neq m\vec{a} + n\vec{b}$ in general.
$\vec{a} = -\vec{b}$ forces $|\vec{a}| = |\vec{b}|$ (magnitudes match even though directions are opposite) $\Rightarrow$ **True**.
$\vec{a} = \vec{b}$ trivially forces equal magnitude $\Rightarrow$ **True**.
Equal magnitude alone says nothing about direction $\Rightarrow$ **False**.
If $\vec{OA} + \vec{AB} + \vec{BC} = \vec{0}$, this says $\vec{OC} = \vec{0}$, i.e. O and C coincide $\Rightarrow$ **True**.
For Q4: if $\vec{a}, \vec{b}$ are collinear, $\vec{b} = \lambda\vec{a}$ for a non-zero scalar λ (True); $\vec{a} = \pm\vec{b}$ only if $|\vec{a}| = |\vec{b}|$, not always (False); components being proportional is exactly what collinearity means (True); direction need not be the same, only along the same/opposite line, and magnitudes can differ (False).

Final Answer

Q3: (i) T (ii) F (iii) F (iv) F (v) F (vi) T (vii) F (viii) T (ix) T (x) F (xi) T (xii) T (xiii) F (xiv) T Q4: (i) T (ii) F (iii) T (iv) F

Common Mistake: Assuming "same magnitude" or "parallel-looking" automatically means "equal vectors" – direction must match exactly, not just the line.

Quick Recall: When in doubt on a T/F vector statement, test it with a quick counterexample (e.g. two same-length vectors pointing in different directions) before deciding.

Type 4

Solving for unknown scalars using the definition of equality of vectors: two vectors are equal only when their corresponding $\hat{i}, \hat{j}, \hat{k}$ components are individually equal.

Formula Used: $\vec{a} = \vec{b} \iff a_1 = b_1,\ a_2 = b_2,\ a_3 = b_3$

Source: Exercise 10.1, Q8

Given: (i) $2\hat{i} + 3\hat{j}$ and $x\hat{i} + y\hat{j}$ are equal, find x, y. (ii) $\vec{a} = x\hat{i} + 2\hat{j} - z\hat{k}$ and $\vec{b} = 3\hat{i} - y\hat{j} + \hat{k}$ are equal, find $x + y + z$. (iii) $\vec{a} = x\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = 3\hat{i} - y\hat{j} + z\hat{k}$, and $2\vec{a} = 3\vec{b}$; find x, y, z.

Solution:

Equate the vectors component-by-component – this is the only rule needed for every part.

Case: Part (i)

Matching $\hat{i}$ and $\hat{j}$ coefficients directly: $x = 2$ and $y = 3$.

Case: Part (ii)

Matching component-wise: $x = 3$ (from $\hat{i}$), $2 = -y \Rightarrow y = -2$ (from $\hat{j}$), $-z = 1 \Rightarrow z = -1$ (from $\hat{k}$). Adding: $x + y + z = 3 - 2 - 1 = 0$.

Case: Part (iii)

First write $2\vec{a} = 2x\hat{i} + 4\hat{j} - 6\hat{k}$ and $3\vec{b} = 9\hat{i} - 3y\hat{j} + 3z\hat{k}$, then equate components.

$2x = 9 \Rightarrow x = \frac{9}{2}$; $\quad 4 = -3y \Rightarrow y = -\frac{4}{3}$; $\quad -6 = 3z \Rightarrow z = -2$.

Final Answer

(i) $x = 2, y = 3$ (ii) $x + y + z = 0$ (iii) $x = \frac{9}{2},\ y = -\frac{4}{3},\ z = -2$

Common Mistake: Forgetting to distribute the scalar (like the 2 and 3 in part iii) into every single component before equating – a very common sign/arithmetic slip.

Quick Recall: Equal vectors mean equal $\hat{i}$, equal $\hat{j}$, equal $\hat{k}$ coefficients – always split into three separate equations first.

Type 5

Finding a vector from its initial and terminal points, and its magnitude. The vector from P to Q is "terminal point minus initial point," and magnitude follows from $\sqrt{\text{sum of squares of components}}$.

Formula Used: $\vec{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

Formula Used: $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

Source: Exercise 10.1, Q12(i)

Given: Initial point $P(2, 3, 0)$, terminal point $Q(-1, -2, -4)$.

Solution:

Subtract the initial point's coordinates from the terminal point's coordinates, component by component, since a vector represents the net displacement from start to end.
$\vec{PQ} = (-1-2)\hat{i} + (-2-3)\hat{j} + (-4-0)\hat{k} = -3\hat{i} - 5\hat{j} - 4\hat{k}$.
Now find the magnitude by taking the square root of the sum of squares of these components.
$|\vec{PQ}| = \sqrt{(-3)^2 + (-5)^2 + (-4)^2} = \sqrt{9+25+16} = \sqrt{50} = 5\sqrt{2}$.

Final Answer

$\vec{PQ} = -3\hat{i} - 5\hat{j} - 4\hat{k}, \quad |\vec{PQ}| = 5\sqrt{2}$

Common Mistake: Subtracting initial minus terminal instead of terminal minus initial – this flips the sign of every component.

Quick Recall: Vector = terminal point − initial point; magnitude = square root of the sum of the squares of its components.

Type 6

Algebraically combining several vectors given in component form – adding, subtracting, and scaling $\hat{i}, \hat{j}, \hat{k}$ vectors term by term.

Source: Exercise 10.1, Q13(i)

Given: $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} + 5\hat{k}$, $\vec{c} = \hat{i} - 6\hat{j} - 7\hat{k}$. Find $\vec{a} + \vec{b} + \vec{c}$.

Solution:

Add the vectors by collecting all the $\hat{i}$ terms together, then all $\hat{j}$ terms, then all $\hat{k}$ terms – vector addition is component-wise.
$\hat{i}$ terms: $1 - 2 + 1 = 0$.
$\hat{j}$ terms: $-2 + 4 - 6 = -4$.
$\hat{k}$ terms: $1 + 5 - 7 = -1$.
So $\vec{a} + \vec{b} + \vec{c} = 0\hat{i} - 4\hat{j} - \hat{k} = -4\hat{j} - \hat{k}$.

Final Answer

$\vec{a} + \vec{b} + \vec{c} = -4\hat{j} - \hat{k}$

Common Mistake: Dropping a component that evaluates to zero (like the $\hat{i}$ term here) instead of writing the final vector correctly without it.

Quick Recall: Add/subtract vectors by combining like components ($\hat{i}$ with $\hat{i}$, $\hat{j}$ with $\hat{j}$, $\hat{k}$ with $\hat{k}$) separately.

Type 7

Finding the unit vector in the direction of a given vector, or in the direction of a sum/difference of two vectors, or along $\vec{AB}$ for given points A, B. Divide the vector by its own magnitude.

★ **Board Favorite** – Unit-vector questions appear in nearly every ISC Vector Algebra paper – a very high-yield, must-know pattern (seen directly, e.g. ISC 2023).

Formula Used: $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$

Source: Exercise 10.1, Q15

Given: $A(1, 2, -3)$ and $B(-1, -2, 1)$ are points of vector $\vec{AB}$. Find the unit vector in the direction of $\vec{AB}$.

Solution:

First build the vector $\vec{AB}$ using terminal minus initial point.
$\vec{AB} = (-1-1)\hat{i} + (-2-2)\hat{j} + (1-(-3))\hat{k} = -2\hat{i} - 4\hat{j} + 4\hat{k}$.
Find its magnitude, since the unit vector requires dividing by this.
$|\vec{AB}| = \sqrt{4+16+16} = \sqrt{36} = 6$.
Divide the vector by its magnitude to get the unit vector.
$\hat{AB} = \dfrac{-2\hat{i} - 4\hat{j} + 4\hat{k}}{6} = \dfrac{1}{3}(-\hat{i} - 2\hat{j} + 2\hat{k})$.

Final Answer

$\hat{AB} = \dfrac{1}{3}(-\hat{i} - 2\hat{j} + 2\hat{k})$

Common Mistake: Computing $\vec{AB}$ as $A - B$ instead of $B - A$, which silently flips the sign of the final unit vector.

Alternate Board-Style Example

Source: Exercise 10.1, Q14(i)

Given: Find the unit vector in the direction of $\vec{a} = 2\hat{i} - 3\hat{j}$.

Solution:

Find the magnitude first: $|\vec{a}| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$.
Divide the vector by this magnitude to normalise it to length 1.
$\hat{a} = \dfrac{2\hat{i} - 3\hat{j}}{\sqrt{13}}$.

Final Answer

$\hat{a} = \dfrac{1}{\sqrt{13}}(2\hat{i} - 3\hat{j})$

Quick Recall: Unit vector = vector ÷ its own magnitude; for $\vec{AB}$, build the vector first (terminal − initial), then normalise.

Type 8

Finding a vector of a specified magnitude that is parallel/collinear to a given vector (or a combination of vectors). Find the unit vector in that direction first, then scale it to the required magnitude.

Formula Used: $\vec{v} = |\vec{v}| \cdot (\widehat{\text{direction}}) = |\vec{v}| \cdot \dfrac{\vec{u}}{|\vec{u}|}$

Source: Exercise 10.1, Q18(iii)

Given: Find all vectors of magnitude $3\sqrt{3}$ which are collinear to $\hat{i} + \hat{j} + \hat{k}$.

Solution:

A vector collinear to $\hat{i} + \hat{j} + \hat{k}$ must be a scalar multiple of it, so first find its unit vector.

$|\hat{i} + \hat{j} + \hat{k}| = \sqrt{1+1+1} = \sqrt{3}$, so the unit vector is $\dfrac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$.

Scale this unit vector to the required magnitude $3\sqrt{3}$; since "collinear" allows either sense along the line, keep both signs.

$\vec{v} = \pm 3\sqrt{3} \cdot \dfrac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}} = \pm 3(\hat{i} + \hat{j} + \hat{k})$.

Final Answer

$\vec{v} = \pm 3(\hat{i} + \hat{j} + \hat{k})$

Common Mistake: Giving only the + sign answer – "collinear" (unlike "in the direction of") includes both directions along the line, so both signs are valid unless the question restricts direction.

Quick Recall: To build a vector of a given magnitude along a direction: find the unit vector, then multiply by the required magnitude (keep ± for "collinear/parallel," drop it for "in the direction of").

Type 9

Finding the direction cosines of a vector, and hence the angles it makes with the coordinate axes. Direction ratios come straight from the components; direction cosines are direction ratios divided by the magnitude.

Formula Used: Direction cosines $(l, m, n) = \left(\frac{a_1}{|\vec{a}|}, \frac{a_2}{|\vec{a}|}, \frac{a_3}{|\vec{a}|}\right)$

Source: Exercise 10.1, Q21(i)

Given: Vector $\hat{i}+\hat{j}-2\hat{k}$. Write its direction ratios and hence find its direction cosines.

Solution:

The direction ratios are simply the components of the vector themselves.
Direction ratios $= \langle 1, 1, -2\rangle$.
Divide each ratio by the magnitude of the vector to normalise them into direction cosines.
$|\vec{a}| = \sqrt{1+1+4} = \sqrt{6}$.
Direction cosines $= \left\langle \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \right\rangle$.

Final Answer

Direction ratios: $\langle 1, 1, -2\rangle$; Direction cosines: $\left\langle \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \right\rangle$

Common Mistake: Confusing direction ratios (any proportional set, unnormalised) with direction cosines (must satisfy $l^2+m^2+n^2 = 1$) – always divide by the magnitude to get true direction cosines.

Quick Recall: Direction ratios are just the raw components; divide each by the magnitude to get direction cosines, which always square-sum to 1.

Type 10

Resolving a vector of known magnitude and known angle (with a stated quadrant) into its x- and y-axis components.

Formula Used: $v_x = |\vec{v}|\cos\theta, \quad v_y = |\vec{v}|\sin\theta$

Source: Exercise 10.1, Q31

Given: $\vec{PQ}$ has magnitude 8 units, makes an angle of 30° with the x-axis, and lies in the fourth quadrant.

Solution:

In the fourth quadrant, x is positive and y is negative, so the angle measured from the positive x-axis (going clockwise) is effectively $-30°$.

Use the standard resolution formulas with $\theta = -30°$ (equivalently, take $\cos 30°$ for x and $-\sin 30°$ for y).

$v_x = 8\cos 30° = 8 \cdot \frac{\sqrt{3}}{2} = 4\sqrt{3}$.

$v_y = -8\sin 30° = -8 \cdot \frac{1}{2} = -4$.

Final Answer

Components: $4\sqrt{3}$ along x-axis, -4 along y-axis

Common Mistake: Forgetting to make the y-component negative for a vector in the fourth quadrant – the magnitude of $\sin 30°$ is used, but the sign must reflect the quadrant.

Quick Recall: Resolve using $|\vec{v}|\cos\theta$ and $|\vec{v}|\sin\theta$, then fix signs according to which quadrant the vector actually lies in.

II. Position Vectors & Section Formula

Type 11

Finding an unknown coordinate of a point when the magnitude of its position vector is given. Set up the magnitude equation and solve for the unknown.

Source: Exercise 10.1, Q19(i)

Given: The magnitude of the position vector of the point $(3, -2, \alpha)$ is 7 units. Find all possible values of α.

Solution:

Write the magnitude-squared equation using all three coordinates, since magnitude is given.
$3^2 + (-2)^2 + \alpha^2 = 7^2 \Rightarrow 9 + 4 + \alpha^2 = 49.$
Isolate α^2 and solve, remembering a square root gives two values.
$\alpha^2 = 36 \Rightarrow \alpha = \pm 6.$

Final Answer

$\alpha = 6$ or $\alpha = -6$

Common Mistake: Taking only the positive square root and missing the second valid value of α.

Quick Recall: Set up $|\text{p.v.}|^2$ = sum of squares of coordinates, then solve for the unknown – always keep both roots.

Type 12

Using the midpoint formula for position vectors, in both directions: finding the midpoint from two given points, and finding an unknown endpoint when the midpoint is given.

Formula Used: $\vec{M} = \dfrac{\vec{a} + \vec{b}}{2}$

Source: Exercise 10.1, Q22

Given: (a) Find the midpoint of P, Q with p.v. $2\hat{\imath}+3\hat{\jmath}-4\hat{k}$ and $4\hat{\imath}+\hat{\jmath}-2\hat{k}$. (b) The p.v. of the midpoint M of AB is $3\hat{\imath}+4\hat{k}$; if the p.v. of A is $4\hat{\imath}-2\hat{\jmath}+3\hat{k}$, find the p.v. of B.

Solution:

Case: Part (a) – direct midpoint

Add the two position vectors and divide by 2.
$\vec{M} = \dfrac{(2\hat{\imath}+3\hat{\jmath}-4\hat{k})+(4\hat{\imath}+\hat{\jmath}-2\hat{k})}{2} = \dfrac{6\hat{\imath}+4\hat{\jmath}-6\hat{k}}{2} = 3\hat{\imath}+2\hat{\jmath}-3\hat{k}$.

Case: Part (b) – reverse midpoint

Rearrange the midpoint formula to solve for the missing endpoint: since $\vec{M} = \dfrac{\vec{A}+\vec{B}}{2}$, we get $\vec{B} = 2\vec{M}-\vec{A}$.
$\vec{B} = 2(3\hat{\imath}+4\hat{k})-(4\hat{\imath}-2\hat{\jmath}+3\hat{k}) = (6\hat{\imath}+8\hat{k})-(4\hat{\imath}-2\hat{\jmath}+3\hat{k}) = 2\hat{\imath}+2\hat{\jmath}+5\hat{k}$.

Final Answer

(a) $\vec{M} = 3\hat{\imath}+2\hat{\jmath}-3\hat{k}$ (b) $\vec{B} = 2\hat{\imath}+2\hat{\jmath}+5\hat{k}$

Common Mistake: In the reverse case, forgetting to double $\vec{M}$ before subtracting $\vec{A}$ – the formula is $\vec{B} = 2\vec{M}-\vec{A}$, not $\vec{M}-\vec{A}$.

Quick Recall: Midpoint = average of the two position vectors; to reverse it, double the midpoint and subtract the known point.

Type 13

Finding the centroid of a triangle from the position vectors of its three vertices – average all three position vectors.

Formula Used: $\vec{G} = \dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$

Source: Exercise 10.1, Q23

Given: Vertices A, B, C of $\triangle ABC$ have position vectors $\hat{\imath}+2\hat{\jmath}-3\hat{k}$, $-2\hat{\imath}-5\hat{\jmath}+4\hat{k}$, $7\hat{\imath}-4\hat{k}$ respectively.

Solution:

Add the three position vectors component-wise, then divide by 3.
$\hat{\imath}$: $1-2+7=6$; $\hat{\jmath}$: $2-5+0=-3$; $\hat{k}$: $-3+4-4=-3$.
Sum $= 6\hat{\imath}-3\hat{\jmath}-3\hat{k}$.
$\vec{G} = \dfrac{6\hat{\imath}-3\hat{\jmath}-3\hat{k}}{3} = 2\hat{\imath}-\hat{\jmath}-\hat{k}$.

Final Answer
$\vec{G} = 2\hat{\imath} - \hat{\jmath} - \hat{k}$

Common Mistake: Dividing only one component by 3 (or forgetting to divide at all) instead of dividing the whole sum vector.

Quick Recall: Centroid = (sum of the three vertex position vectors) ÷ 3, exactly like averaging coordinates.

Type 14

Using the section formula to find the position vector of a point dividing a segment joining two given points, in a given ratio – internally or externally.

★ **Board Favorite** – Section formula (especially the internal/external contrast) is a recurring, high-yield ISC pattern tested almost every year in some form.

Formula Used: Internal division $(m : n)$: $\vec{R} = \dfrac{m\vec{b} + n\vec{a}}{m+n}$

Formula Used: External division $(m : n)$: $\vec{R} = \dfrac{m\vec{b} - n\vec{a}}{m-n}$

Source: Exercise 10.1, Q24

Given: P, Q have position vectors $\hat{\imath} + 2\hat{\jmath} - \hat{k}$ and $-\hat{\imath} + \hat{\jmath} + \hat{k}$. Find the position vector of R dividing PQ in ratio $2 : 1$ (i) internally (ii) externally.

Solution:

Case: Internal division

Apply the internal section formula with $m = 2, n = 1$ (the ratio is $PR : RQ = 2 : 1$, so Q carries weight m and P carries weight n).

$$\vec{R} = \frac{2(-\hat{\imath} + \hat{\jmath} + \hat{k}) + 1(\hat{\imath} + 2\hat{\jmath} - \hat{k})}{2+1} = \frac{(-2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}) + (\hat{\imath} + 2\hat{\jmath} - \hat{k})}{3} = \frac{-\hat{\imath} + 4\hat{\jmath} + \hat{k}}{3}.$$

Case: External division

Apply the external section formula, which subtracts instead of adds.

$$\vec{R} = \frac{2(-\hat{\imath} + \hat{\jmath} + \hat{k}) - 1(\hat{\imath} + 2\hat{\jmath} - \hat{k})}{2-1} = (-2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}) - (\hat{\imath} + 2\hat{\jmath} - \hat{k}) = -3\hat{\imath} + 3\hat{k}.$$

Final Answer

Internal: $\dfrac{-\hat{i}+4\hat{j}+\hat{k}}{3}$ External: $-3\hat{i}+3\hat{k}$

Common Mistake: Using + in the denominator or numerator for external division – external division always uses a *difference*, both top and bottom.

Alternate Board-Style Example

Source: Exercise 10.1, Q25(i)

Given: Find the position vector of the point dividing the join of $\vec{a}+3\vec{b}$ and $\vec{a}-\vec{b}$ internally in ratio $1:3$.

Solution:

Apply the internal section formula with the two given position vectors as the endpoints and $m=1, n=3$.

$\vec{R} = \dfrac{1\cdot(\vec{a}-\vec{b})+3\cdot(\vec{a}+3\vec{b})}{1+3} = \dfrac{(\vec{a}-\vec{b})+(3\vec{a}+9\vec{b})}{4} = \dfrac{4\vec{a}+8\vec{b}}{4} = \vec{a}+2\vec{b}.$

Final Answer

$\vec{R}=\vec{a}+2\vec{b}$

Quick Recall: Internal division adds weighted position vectors and divides by the sum of the ratio; external division subtracts them and divides by the difference of the ratio.

Type 15

Expressing vectors between named points (like $\vec{AC}, \vec{BD}$) purely in terms of given position vectors $\vec{a}, \vec{b}$, using "terminal minus initial."

Source: Exercise 10.1, Q27

Given: Points A, B, C, D have position vectors $\vec{a}, \vec{b}, 2\vec{a}+3\vec{b}, 2\vec{a}-3\vec{b}$ respectively. Express $\vec{AC}, \vec{BD}, \vec{CD}, \vec{AB}$ in terms of $\vec{a}$ and $\vec{b}$.

Solution:

Apply "terminal point's p.v. minus initial point's p.v." to each requested vector in turn.

$\vec{AC} = (2\vec{a}+3\vec{b})-\vec{a} = \vec{a}+3\vec{b}.$

$\vec{BD} = (2\vec{a}-3\vec{b})-\vec{b} = 2\vec{a}-4\vec{b}.$

$\vec{CD} = (2\vec{a}-3\vec{b})-(2\vec{a}+3\vec{b}) = -6\vec{b}.$

$\vec{AB} = \vec{b}-\vec{a}.$

Final Answer

$\vec{AC} = \vec{a} + 3\vec{b}$, $\vec{BD} = 2\vec{a} - 4\vec{b}$, $\vec{CD} = -6\vec{b}$, $\vec{AB} = \vec{b} - \vec{a}$

Common Mistake: Reversing the order (writing $\vec{AC}$ as $\vec{a} - (2\vec{a} + 3\vec{b})$) – always terminal minus initial, matching the order of the letters in the vector's name.

Quick Recall: For any $\vec{XY}$ in terms of position vectors, compute (p.v. of Y) − (p.v. of X) – the letter order tells you the order of subtraction.

III. Collinearity & Vector Proofs of Figures

Type 16

Proving three points are collinear using the vector method: show that the vector joining two of the points is a scalar multiple of the vector joining another pair sharing a common point.

★ **Board Favorite** – Vector-method collinearity proofs are a standard, frequently tested ISC long-answer pattern (appeared in ISC 2024).

Formula Used: A, B, C collinear $\iff$ $\vec{AB} = \lambda\,\vec{AC}$ for some scalar λ

Source: Exercise 10.1, Q32(i)

Given: Points $(2, -1, 3)$, $(3, -5, 1)$, $(-1, 11, 9)$. Prove they are collinear using vectors.

Solution:

Let $A(2, -1, 3)$, $B(3, -5, 1)$, $C(-1, 11, 9)$. Find two vectors that share the common point A.

$\vec{AB} = (3-2)\hat{\imath} + (-5+1)\hat{\jmath} + (1-3)\hat{k} = \hat{\imath} - 4\hat{\jmath} - 2\hat{k}$.

$\vec{AC} = (-1-2)\hat{\imath} + (11+1)\hat{\jmath} + (9-3)\hat{k} = -3\hat{\imath} + 12\hat{\jmath} + 6\hat{k}$.

Check whether $\vec{AC}$ is a scalar multiple of $\vec{AB}$, since that would mean they lie along the same line.

$\vec{AC} = -3(\hat{\imath} - 4\hat{\jmath} - 2\hat{k}) = -3\,\vec{AB}$.

Since $\vec{AC} = -3\vec{AB}$, the vectors are parallel; and since they also share the common point A, the points A, B, C must be collinear.

Final Answer

$\vec{AC} = -3\vec{AB}$, so A, B, C are collinear.

Common Mistake: Showing the vectors are parallel but forgetting to state that they also share a common point – both conditions together are what prove collinearity, not parallelism alone.

Alternate Board-Style Example

Source: Exercise 10.1, Q34(i)

Given: If the points $(\alpha, -1)$, $(2, 1)$, $(4, 5)$ are collinear, find α by the vector method.

Solution:

Let $A(\alpha,-1)$, $B(2,1)$, $C(4,5)$. Find $\vec{AB}$ and $\vec{BC}$, which share the common point B.
$\vec{AB}=(2-\alpha)\hat{\imath}+2\hat{\jmath}, \quad \vec{BC}=2\hat{\imath}+4\hat{\jmath}$.
For collinearity, $\vec{AB}$ must be a scalar multiple of $\vec{BC}$, so their components must be proportional.
$\frac{2-\alpha}{2}=\frac{2}{4}=\frac{1}{2}\Rightarrow 2-\alpha=1\Rightarrow\alpha=1$.

Final Answer

$\alpha=1$

Quick Recall: To prove collinearity, show that a vector between two of the points is a scalar multiple of a vector between another pair sharing a common point.

Type 17

Finding an unknown parameter that makes two given vectors parallel (a special/degenerate case of the collinearity idea) – match corresponding components as a proportion.

Source: Exercise 10.1, Q20(ii)

Given: Find the value of p for which $\vec{a}=3\hat{\imath}+2\hat{\jmath}+9\hat{k}$ and $\vec{b}=\hat{\imath}+p\hat{\jmath}+3\hat{k}$ are parallel vectors.

Solution:

For two vectors to be parallel, their corresponding components must all be in the same ratio.
$\frac{3}{1}=\frac{2}{p}=\frac{9}{3}$.
The first and third ratios both equal 3, confirming consistency; use this common ratio with the middle term to solve for p.
$\frac{2}{p}=3\Rightarrow p=\frac{2}{3}$.

Final Answer

$p=\frac{2}{3}$

Common Mistake: Cross-checking only one pair of components and not verifying that all three ratios are mutually consistent before solving.

Quick Recall: Parallel vectors have proportional components – set all corresponding ratios equal and solve for the unknown.

Type 18

Using the vector method to prove a triangle (given by position vectors of its vertices) is right-angled, equilateral, or isosceles – compute the three side vectors and compare their magnitudes/dot products.

Source: Exercise 10.1, Q36(i)

Given: Points with position vectors $3\hat{\imath} - 4\hat{\jmath} - 4\hat{k}$, $2\hat{\imath} - \hat{\jmath} + \hat{k}$, $\hat{\imath} - 3\hat{\jmath} - 5\hat{k}$ form the vertices of a triangle. Show it is right-angled. (NCERT)

Solution:

Let A, B, C be the three points in the given order. Find all three side vectors.
$\vec{AB} = B - A = -\hat{\imath} + 3\hat{\jmath} + 5\hat{k}$, $\quad \vec{BC} = C - B = -\hat{\imath} - 2\hat{\jmath} - 6\hat{k}$, $\quad \vec{CA} = A - C = 2\hat{\imath} - \hat{\jmath} + \hat{k}$.
Compute the square of each side's length, since Pythagoras' theorem is the cleanest right-angle test here.
$|\vec{AB}|^2 = 1 + 9 + 25 = 35$, $\quad |\vec{BC}|^2 = 1 + 4 + 36 = 41$, $\quad |\vec{CA}|^2 = 4 + 1 + 1 = 6$.
Check whether the sum of the two smaller squares equals the largest square.
$|\vec{AB}|^2 + |\vec{CA}|^2 = 35 + 6 = 41 = |\vec{BC}|^2$.
Since this matches Pythagoras' theorem with BC as the hypotenuse, the triangle is right-angled at A (the vertex common to the two legs AB and CA).

Final Answer

$\triangle ABC$ is right-angled at A.

Common Mistake: Identifying the right angle at the wrong vertex – the right angle sits at the vertex shared by the two *legs* (the shorter sides), not at an endpoint of the hypotenuse.

Quick Recall: Find all three side-vector magnitudes squared; if two of them add up to the third, it's a right triangle, right-angled at their shared vertex.

Type 19

Using vector addition in a parallelogram to find the magnitudes of its diagonals from two given adjacent side vectors.

Formula Used: One diagonal $= \vec{PQ} + \vec{PS}$, $\qquad$ Other diagonal $= \vec{PS} - \vec{PQ}$

Source: Exercise 10.1, Q28

Given: In parallelogram $PQRS$, $\vec{PQ} = 3\hat{\imath} - 2\hat{\jmath} + 2\hat{k}$ and $\vec{PS} = -\hat{\imath} - 2\hat{k}$. Find $|\vec{PR}|$ and $|\vec{QS}|$.

Solution:

The diagonal PR goes from P straight to the opposite vertex R, which is reached by travelling along PQ then QR (and $QR = PS$ in a parallelogram), so $\vec{PR} = \vec{PQ} + \vec{PS}$.
$\vec{PR} = (3\hat{i} - 2\hat{j} + 2\hat{k}) + (-\hat{i} - 2\hat{k}) = 2\hat{i} - 2\hat{j}$.
$|\vec{PR}| = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$.
The other diagonal QS goes from Q to S, i.e. $\vec{QS} = \vec{PS} - \vec{PQ}$ (subtracting the position of Q relative to P from that of S).
$\vec{QS} = (-\hat{i} - 2\hat{k}) - (3\hat{i} - 2\hat{j} + 2\hat{k}) = -4\hat{i} + 2\hat{j} - 4\hat{k}$.
$|\vec{QS}| = \sqrt{16+4+16} = \sqrt{36} = 6$.

Final Answer

$|\vec{PR}| = 2\sqrt{2}, \quad |\vec{QS}| = 6$

Common Mistake: Mixing up which combination (sum or difference of the two side vectors) gives which diagonal – always sketch the parallelogram quickly to check.

Quick Recall: In a parallelogram with adjacent sides $\vec{PQ}, \vec{PS}$ from the same vertex: one diagonal is their sum, the other is their difference.

IV. Scalar (Dot) Product

Type 20

Expanding a dot-product identity like $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})$ using distributivity to isolate an unknown magnitude, given that one vector is a unit vector.

Formula Used: $\vec{a}\cdot\vec{a}=|\vec{a}|^2$, dot product is distributive over addition

Source: Exercise 10.2, Q1(i)

Given: $\vec{a}$ is a unit vector and $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=15$. Find $|\vec{x}|$.

Solution:

Expand the product just like $(m-n)(m+n)=m^2-n^2$, since dot product distributes the same way.
$(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=\vec{x}\cdot\vec{x}+\vec{x}\cdot\vec{a}-\vec{a}\cdot\vec{x}-\vec{a}\cdot\vec{a}=\vec{x}\cdot\vec{x}-\vec{a}\cdot\vec{a}=|\vec{x}|^2-|\vec{a}|^2$.
Since $\vec{a}$ is a unit vector, $|\vec{a}|^2=1$; substitute the given value.
$|\vec{x}|^2-1=15\Rightarrow|\vec{x}|^2=16\Rightarrow|\vec{x}|=4$.

Final Answer

$|\vec{x}|=4$

Common Mistake: Forgetting that $\vec{x}\cdot\vec{a}=\vec{a}\cdot\vec{x}$ (dot product is commutative), so these cross terms should cancel rather than double up.

Quick Recall: Dot-product expansions behave like ordinary algebra (a^2-b^2 style); a unit vector always contributes 1 when dotted with itself.

Type 21

Finding the angle between two vectors, either directly from given magnitudes and dot product, or after first computing the dot product from components.

★ **Board Favorite** – "Find the angle between two vectors" is one of the most repeated ISC question stems for this chapter, in both magnitude-given and component forms.

Formula Used: $\cos\theta=\dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

Source: Exercise 10.2, Q4(i)

Given: $|\vec{a}| = 2,\ |\vec{b}| = \sqrt{3},\ \vec{a}\cdot\vec{b} = 3$. Find the angle between $\vec{a}$ and $\vec{b}$.

Solution:

Substitute directly into the angle formula, since the magnitudes and dot product are already given.
$\cos\theta = \dfrac{3}{2\cdot\sqrt{3}} = \dfrac{3}{2\sqrt{3}} = \dfrac{\sqrt{3}}{2}$.
Recognise this as a standard cosine value.
$\theta = 30^\circ$.

Final Answer

$\theta = 30^\circ$

Common Mistake: Leaving the answer as $\dfrac{3}{2\sqrt{3}}$ without rationalising/simplifying to the recognisable standard value $\dfrac{\sqrt{3}}{2}$.

Alternate Board-Style Example

Source: Exercise 10.2, Q5 (Exemplar)

Given: Find the angle between the vectors $2\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} + 4\hat{j} - \hat{k}$.

Solution:

Compute the dot product in component form first.
$\vec{a}\cdot\vec{b} = (2)(3) + (-1)(4) + (1)(-1) = 6 - 4 - 1 = 1$.
Find each magnitude separately.
$|\vec{a}| = \sqrt{4+1+1} = \sqrt{6}, \quad |\vec{b}| = \sqrt{9+16+1} = \sqrt{26}$.
Substitute into the angle formula.
$\cos\theta = \dfrac{1}{\sqrt{6}\cdot\sqrt{26}} = \dfrac{1}{\sqrt{156}}$.

Final Answer

$\theta = \cos^{-1}\left(\dfrac{1}{\sqrt{156}}\right)$

Quick Recall: Angle between vectors always comes from $\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$ – compute dot product and both magnitudes, then substitute.

Type 22

Finding the common magnitude of two vectors that have equal length, a given angle between them, and a given scalar product.

Source: Exercise 10.2, Q3

Given: Two vectors $\vec{a}, \vec{b}$ have the same magnitude, the angle between them is $60°$, and their scalar product is $\frac{9}{2}$.

Solution:

Let the common magnitude be x, so $|\vec{a}| = |\vec{b}| = x$; substitute into the dot-product formula.
$\vec{a} \cdot \vec{b} = x \cdot x \cdot \cos 60° = x^2 \cdot \frac{1}{2} = \frac{x^2}{2}$.
Set this equal to the given scalar product and solve for x.
$\frac{x^2}{2} = \frac{9}{2} \Rightarrow x^2 = 9 \Rightarrow x = 3$ (magnitude must be positive).

Final Answer

$|\vec{a}| = |\vec{b}| = 3$

Common Mistake: Keeping the negative root $x = -3$ – magnitude can never be negative, so only the positive root is valid.

Quick Recall: With equal magnitudes x, $\vec{a} \cdot \vec{b} = x^2 \cos\theta$ – solve this single equation for x.

Type 23

Computing the dot product of linear combinations of vectors given in component form – simplify each combination first, then take the dot product.

Source: Exercise 10.2, Q6(i)

Given: $\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$. Find $(\vec{b} - \vec{a}) \cdot (3\vec{a} + \vec{b})$.

Solution:

Simplify each bracket separately in component form before dotting them.
$\vec{b} - \vec{a} = (2-1)\hat{i} + (1+2)\hat{j} + (-3-5)\hat{k} = \hat{i} + 3\hat{j} - 8\hat{k}$.
$3\vec{a} + \vec{b} = (3+2)\hat{i} + (-6+1)\hat{j} + (15-3)\hat{k} = 5\hat{i} - 5\hat{j} + 12\hat{k}$.
Now take the dot product of the two simplified vectors, component by component.
$(\hat{i} + 3\hat{j} - 8\hat{k}) \cdot (5\hat{i} - 5\hat{j} + 12\hat{k}) = (1)(5) + (3)(-5) + (-8)(12) = 5 - 15 - 96 = -106$.

Final Answer

-106

Common Mistake: Trying to dot the original unsimplified expressions term by term without first collecting each bracket – this multiplies the chance of a sign error.

Quick Recall: Always simplify each bracket into a single vector first, then dot the two resulting vectors component-wise.

Type 24

Finding the (scalar) projection of one vector on another – either computing it directly, or reverse-solving for an unknown component/parameter when the projection value is given.

Formula Used: Projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

Source: Exercise 10.2, Q7(i), Q7(iv)

Given: (i) $\vec{a}\cdot\vec{b} = 8$, $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$; find the projection of $\vec{a}$ on $\vec{b}$. (iv) Find λ when the projection of $\hat{i} + \lambda\hat{j} + \hat{k}$ on $\hat{i} + \hat{j}$ is $\sqrt{2}$.

Solution:

Case: Part (i) – direct projection

Find $|\vec{b}|$ first, then divide the given dot product by it.
$|\vec{b}| = \sqrt{4 + 36 + 9} = \sqrt{49} = 7$.
Projection $= \dfrac{8}{7}$.

Case: Part (iv) – reverse-solve for λ

Write the projection formula symbolically and set it equal to the given value.
$(\hat{i} + \lambda\hat{j} + \hat{k})\cdot(\hat{i} + \hat{j}) = 1 + \lambda$, and $|\hat{i} + \hat{j}| = \sqrt{2}$.
$\dfrac{1+\lambda}{\sqrt{2}} = \sqrt{2} \Rightarrow 1 + \lambda = 2 \Rightarrow \lambda = 1$.

Final Answer

(i) Projection $= \dfrac{8}{7}$ (iv) $\lambda = 1$

Common Mistake: Dividing by $|\vec{a}|$ instead of $|\vec{b}|$ – the projection of $\vec{a}$ *on* $\vec{b}$ always divides by the magnitude of the vector being projected *onto*.

Quick Recall: Projection of $\vec{a}$ on $\vec{b}$ is $\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$ – for reverse problems, just set this expression equal to the given projection value and solve.

Type 25

Finding an unknown scalar that makes two vectors perpendicular, and proving perpendicularity identities – both rest on setting the dot product to zero.

Formula Used: $\vec{a} \perp \vec{b} \iff \vec{a}\cdot\vec{b} = 0$

Source: Exercise 10.2, Q8(ii)

Given: $\vec{a} = 3\hat{i} - \hat{j} + 4\hat{k}$ and $\vec{b} = -\lambda\hat{i} + 3\hat{j} + 3\hat{k}$ are perpendicular. Find λ.

Solution:

Set the dot product of the two vectors equal to zero, since that is exactly the condition for perpendicularity.

$\vec{a}\cdot\vec{b} = (3)(-\lambda) + (-1)(3) + (4)(3) = -3\lambda - 3 + 12 = -3\lambda + 9.$

Solve the resulting linear equation for λ.

$-3\lambda + 9 = 0 \Rightarrow \lambda = 3.$

Final Answer

$\lambda = 3$

Common Mistake: Sign errors while distributing the dot product across each term – write out each product carefully rather than doing it mentally.

Quick Recall: Perpendicular vectors always satisfy $\vec{a}\cdot\vec{b} = 0$ – set up this single equation and solve for the unknown.

Type 26

Deriving identities from a vector sum condition like $\vec{a}+\vec{b}+\vec{c} = \vec{0}$ (unit vectors), and from $|\vec{a}\pm\vec{b}|^2$ expansions – square both sides and expand using the dot product.

Formula Used: $|\vec{a}+\vec{b}+\vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

Source: Exercise 10.2, Q12(i), Q10

Given: (i) $\vec{a}, \vec{b}, \vec{c}$ are unit vectors with $\vec{a}+\vec{b}+\vec{c} = \vec{0}$; find $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$. Also: if $\vec{a}, \vec{b}$ are unit vectors, $c = |\vec{a}+\vec{b}|$, $d = |\vec{a}-\vec{b}|$, find $c^2 + d^2$.

Solution:

Case: Part (i)

Square both sides of $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, i.e. take the dot product of the sum with itself, since squaring a zero vector gives zero.
$|\vec{a}+\vec{b}+\vec{c}|^2=0 \Rightarrow |\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0.$
Substitute $|\vec{a}|^2=|\vec{b}|^2=|\vec{c}|^2=1$ (unit vectors) and solve.
$3+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0 \Rightarrow \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-\frac{3}{2}.$

Case: Second identity (c^2+d^2)

Expand $|\vec{a}+\vec{b}|^2$ and $|\vec{a}-\vec{b}|^2$ separately using the dot product, then add them – the cross terms $2\vec{a}\cdot\vec{b}$ cancel.
$c^2+d^2=(|\vec{a}|^2+2\vec{a}\cdot\vec{b}+|\vec{b}|^2)+(|\vec{a}|^2-2\vec{a}\cdot\vec{b}+|\vec{b}|^2)=2|\vec{a}|^2+2|\vec{b}|^2=2(1)+2(1)=4.$

Final Answer

$\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-\frac{3}{2};\qquad c^2+d^2=4$

Common Mistake: Forgetting the factor of 2 in front of the cross terms when expanding $|\vec{a}+\vec{b}+\vec{c}|^2$ – there are three distinct cross-term pairs, each doubled.

Quick Recall: "Square" any vector sum/difference condition using the dot product with itself – this is the standard trick whenever a sum of vectors equals zero or a fixed magnitude.

Type 27

Short conceptual/justification questions on dot-product properties – test a claim with a quick counterexample or recall the exact condition.

Source: Exercise 10.2, Q13(i), Q13(ii)

Given: (i) Given $(\vec{a})^2=(\vec{b})^2$, is it necessary that $\vec{a}=\vec{b}$? Justify. (ii) For non-zero $\vec{a},\vec{b}$, when does $|\vec{a}+\vec{b}|^2=|\vec{a}|^2+|\vec{b}|^2$ hold?

Solution:

For part (i), test with a simple counterexample rather than assuming equality of magnitudes forces equality of vectors.
Take $\vec{a}=2\hat{i}+\hat{j}$ and $\vec{b}=\hat{i}-2\hat{j}$: both have magnitude $\sqrt{5}$, so $(\vec{a})^2=(\vec{b})^2$, yet $\vec{a}\neq\vec{b}$. So it is **not necessary**.
For part (ii), expand $|\vec{a}+\vec{b}|^2$ and compare to the right-hand side.
$|\vec{a}+\vec{b}|^2=|\vec{a}|^2+2\vec{a}\cdot\vec{b}+|\vec{b}|^2$. This equals $|\vec{a}|^2+|\vec{b}|^2$ only when the cross term vanishes, i.e. $\vec{a}\cdot\vec{b}=0$, meaning $\vec{a}\perp\vec{b}$.

Final Answer

(i) No – equal magnitude does not force equal vectors. (ii) Holds exactly when $\vec{a} \perp \vec{b}$.

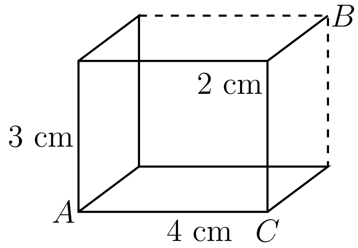
Common Mistake: Confusing "equal magnitude" with "equal vector" – magnitude alone never determines direction.

Quick Recall: $(\vec{a})^2 = (\vec{b})^2$ only compares lengths, not direction; $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2$ is exactly the vector version of the Pythagoras theorem, i.e. perpendicularity.

Type 28

Applying the dot product to a 3D solid (a cuboid) with edges given along the figure – set up coordinates matching the labelled edges, then compute the dot product of the requested vectors.

Source: Exercise 10.2, Q14(iii) (ISC Specimen Paper 2025)



Cuboid with edges 2 cm, 3 cm, 4 cm meeting near vertex A; B is a top vertex, C a base vertex.

Given: A cuboid with edges 2 cm, 3 cm, 4 cm labelled as shown, with vertices A, B, C marked on the figure.

Solution:

Set up coordinates with A at the origin, using the three mutually perpendicular edges of the cuboid as the axes.

Take $A = (0,0,0)$; going 4 cm along the base gives $C = (4,0,0)$; going 3 cm up and then 2 cm along the depth gives $B = (0,2,3)$ (matching the edges as labelled in the figure).

Build the two vectors needed, both starting from B.

$\vec{BA} = A - B = (0,-2,-3)$, $\quad \vec{BC} = C - B = (4,-2,-3)$.

Take the dot product component-wise.

$\vec{BA} \cdot \vec{BC} = (0)(4) + (-2)(-2) + (-3)(-3) = 0 + 4 + 9 = 13.$

Final Answer

$\vec{BA} \cdot \vec{BC} = 13$ (with A, B, C placed as coordinates $(0,0,0), (0,2,3), (4,0,0)$ respectively – *Adnan: please cross-check this vertex correspondence against the exact printed figure, since the book's answer key lists* 11*; the value depends on precisely which vertices* B *and the hidden* 2 *cm edge connect to.*)

Common Mistake: Assigning the wrong vertex as the "depth" corner – always match each labelled edge in the figure to a specific coordinate axis before writing any vector.

Quick Recall: For solid-figure dot-product questions: place one vertex at the origin, use the three edge lengths as the three axes, then subtract coordinates to build each vector.

Type 29

Solving a compound dot-product equation (a linear combination of two vectors dotted with another combination) to find an unknown magnitude, using a stated perpendicularity condition.

Source: Exercise 10.2, Q14(ii)

Given: $\vec{a}$ is a unit vector perpendicular to $\vec{b}$, and $(\vec{a} + 2\vec{b}) \cdot (3\vec{a} - \vec{b}) = -5$. Find $|\vec{b}|$.

Solution:

Expand the product using distributivity, just like expanding $(x + 2y)(3x - y)$ in ordinary algebra.
$(\vec{a} + 2\vec{b}) \cdot (3\vec{a} - \vec{b}) = 3\vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} + 6\vec{b} \cdot \vec{a} - 2\vec{b} \cdot \vec{b} = 3|\vec{a}|^2 + 5(\vec{a} \cdot \vec{b}) - 2|\vec{b}|^2$.
Use the given facts: $\vec{a}$ is a unit vector ($|\vec{a}|^2 = 1$) and $\vec{a} \perp \vec{b}$ ($\vec{a} \cdot \vec{b} = 0$), then substitute.
$3(1) + 5(0) - 2|\vec{b}|^2 = -5 \Rightarrow 3 - 2|\vec{b}|^2 = -5$.
Solve for $|\vec{b}|^2$, then take the square root.
$2|\vec{b}|^2 = 8 \Rightarrow |\vec{b}|^2 = 4 \Rightarrow |\vec{b}| = 2$.

Final Answer

$|\vec{b}| = 2$

Common Mistake: Forgetting to combine the two middle cross terms ($-\vec{a} \cdot \vec{b}$ and $6\vec{b} \cdot \vec{a}$) into $5(\vec{a} \cdot \vec{b})$ before applying the perpendicularity condition.

Quick Recall: Expand compound dot products exactly like algebraic brackets, then plug in known facts (unit vector $\Rightarrow$ self-dot is 1; perpendicular $\Rightarrow$ cross-dot is 0).

Type 30

Verifying that a given set of vectors are mutually perpendicular unit vectors – check every pairwise dot product is zero, and every vector's magnitude is 1.

Source: Exercise 10.2, Q22 (NCERT)

Given: Show that $\frac{1}{7}(2\hat{i}+3\hat{j}+6\hat{k})$, $\frac{1}{7}(3\hat{i}-6\hat{j}+2\hat{k})$, $\frac{1}{7}(6\hat{i}+2\hat{j}-3\hat{k})$ are mutually perpendicular unit vectors.

Solution:

Call them $\vec{p}, \vec{q}, \vec{r}$. First check that each has magnitude 1, since a unit vector requirement must be checked separately from perpendicularity.

$|\vec{p}|^2 = \frac{1}{49}(4+9+36) = \frac{49}{49} = 1$, and similarly $|\vec{q}|^2 = \frac{1}{49}(9+36+4) = 1$, $|\vec{r}|^2 = \frac{1}{49}(36+4+9) = 1$. So all three are unit vectors.

Now check every pairwise dot product is zero, since that confirms mutual perpendicularity.

$\vec{p}\cdot\vec{q} = \frac{1}{49}\big[(2)(3)+(3)(-6)+(6)(2)\big] = \frac{1}{49}(6-18+12) = 0.$

$\vec{q}\cdot\vec{r} = \frac{1}{49}\big[(3)(6)+(-6)(2)+(2)(-3)\big] = \frac{1}{49}(18-12-6) = 0.$

$\vec{r}\cdot\vec{p} = \frac{1}{49}\big[(6)(2)+(2)(3)+(-3)(6)\big] = \frac{1}{49}(12+6-18) = 0.$

All three pairwise dot products vanish and all three magnitudes equal 1, so the vectors are indeed mutually perpendicular unit vectors.

Final Answer

Verified: all magnitudes $= 1$ and all pairwise dot products $= 0$.

Common Mistake: Checking only one pair of vectors for perpendicularity and assuming the third pair follows automatically – all three pairs must be checked independently.

Quick Recall: "Mutually perpendicular unit vectors" needs two separate checks for every vector/pair: magnitude $= 1$ each, and dot product $= 0$ for every pair.

Type 31

Proving the identity $\cos\frac{\theta}{2} = \frac{1}{2}|\vec{a}+\vec{b}|$ for two unit vectors with angle θ between them – expand $|\vec{a}+\vec{b}|^2$ and use the half-angle cosine identity.

Formula Used: $1+\cos\theta = 2\cos^2\frac{\theta}{2}$

Source: Exercise 10.2, Q23

Given: $\vec{a}, \vec{b}$ are unit vectors and θ is the angle between them. Prove $\cos\frac{\theta}{2} = \frac{1}{2}|\vec{a}+\vec{b}|$.

Solution:

Start from $|\vec{a}+\vec{b}|^2$ and expand using the dot product.
$|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + |\vec{b}|^2 = 1 + 2\cos\theta + 1 = 2 + 2\cos\theta = 2(1+\cos\theta)$ (using $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = \cos\theta$ since both are unit vectors).
Apply the half-angle identity $1+\cos\theta = 2\cos^2\frac{\theta}{2}$ to rewrite the right-hand side.
$|\vec{a}+\vec{b}|^2 = 2\cdot 2\cos^2\frac{\theta}{2} = 4\cos^2\frac{\theta}{2}$.
Take the square root of both sides (cosine of a half-angle between 0 and π is non-negative, so no sign ambiguity).
$|\vec{a}+\vec{b}| = 2\cos\frac{\theta}{2} \Rightarrow \cos\frac{\theta}{2} = \frac{1}{2}|\vec{a}+\vec{b}|$.

Final Answer

Proved: $\cos\frac{\theta}{2} = \frac{1}{2}|\vec{a}+\vec{b}|$

Common Mistake: Forgetting that $\theta \in [0,\pi]$ makes $\theta/2 \in [0,\pi/2]$, where cosine is always non-negative – this is why the square root can be taken without a $\pm$ sign.

Quick Recall: This proof is just "expand $|\vec{a}+\vec{b}|^2$ with the dot product, then recognise the half-angle identity $1+\cos\theta = 2\cos^2(\theta/2)$."

Type 32

Finding the angles a vector makes with each of the three coordinate axes, using the dot product with $\hat{i}, \hat{j}, \hat{k}$ (equivalently, the direction cosines).

Formula Used: $\cos\alpha = \frac{\vec{a}\cdot\hat{i}}{|\vec{a}|}, \quad \cos\beta = \frac{\vec{a}\cdot\hat{j}}{|\vec{a}|}, \quad \cos\gamma = \frac{\vec{a}\cdot\hat{k}}{|\vec{a}|}$

Source: Exercise 10.2, Q24

Given: Find the angles which the vector $\vec{a} = 3\hat{i} - 6\hat{j} + 2\hat{k}$ makes with the coordinate axes.

Solution:

Find the magnitude of the vector first, since it appears in every direction cosine.
$|\vec{a}| = \sqrt{9+36+4} = \sqrt{49} = 7$.
The dot product of $\vec{a}$ with each unit axis vector just picks out that component, so the direction cosines are the components divided by the magnitude.
$\cos\alpha = \frac{3}{7}, \quad \cos\beta = \frac{-6}{7}, \quad \cos\gamma = \frac{2}{7}$.

Final Answer

$\alpha = \cos^{-1}\left(\frac{3}{7}\right)$, $\beta = \cos^{-1}\left(\frac{-6}{7}\right)$, $\gamma = \cos^{-1}\left(\frac{2}{7}\right)$

Common Mistake: Dropping the negative sign on $\cos\beta$ – a negative direction cosine is perfectly valid and means the angle with that axis is obtuse.

Quick Recall: The angle a vector makes with an axis has cosine equal to (component along that axis) ÷ (magnitude of the vector).

Type 33

Finding the range of a parameter for which the angle between two vectors is obtuse – an obtuse angle needs a strictly negative dot product (and the vectors not exactly anti-parallel).

Formula Used: θ is obtuse $\iff$ $\vec{a} \cdot \vec{b} < 0$ (and $\vec{a}, \vec{b}$ not anti-parallel)

Source: Exercise 10.2, Q25 (ISC Specimen Paper 2025)

Given: Find the values of x for which the angle between the vectors $2x^2\hat{i} + 3x\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + x^2\hat{k}$ is obtuse.

Solution:

Compute the dot product of the two vectors in terms of x.
$\vec{a} \cdot \vec{b} = (2x^2)(1) + (3x)(-2) + (1)(x^2) = 2x^2 - 6x + x^2 = 3x^2 - 6x$.
For an obtuse angle, this dot product must be strictly negative.
$3x^2 - 6x < 0 \Rightarrow 3x(x - 2) < 0$.
Solve this quadratic inequality – the expression is negative between its roots 0 and 2.
$0 < x < 2$.

Final Answer

$x \in (0, 2)$

Common Mistake: Including $x = 0$ or $x = 2$ in the answer – at these endpoints the dot product is exactly zero (a right angle, not obtuse), so the interval must be strictly open.

Quick Recall: Obtuse angle $\iff$ dot product < 0 – turn the dot product into an inequality in the unknown and solve, keeping the inequality strict.

Type 34

Finding an unknown vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ from its dot products with three given vectors – this produces a system of three linear equations in x, y, z.

Source: Exercise 10.2, Q27

Given: Dot products of a vector with $3\hat{i} - 5\hat{k}$, $2\hat{i} + 7\hat{j}$ and $\hat{i} + \hat{j} + \hat{k}$ are respectively -1, 6, 5. Find the vector.

Solution:

Let the required vector be $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, and translate each dot-product condition into a linear equation.
$\vec{r} \cdot (3\hat{i} - 5\hat{k}) = -1 \Rightarrow 3x - 5z = -1 \quad \ldots(1)$
$\vec{r} \cdot (2\hat{i} + 7\hat{j}) = 6 \Rightarrow 2x + 7y = 6 \quad \ldots(2)$
$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 5 \Rightarrow x + y + z = 5 \quad \ldots(3)$
From (3), express z in terms of x, y and substitute into (1) to eliminate z.
$z = 5 - x - y \Rightarrow 3x - 5(5 - x - y) = -1 \Rightarrow 3x - 25 + 5x + 5y = -1 \Rightarrow 8x + 5y = 24$ $\ldots(4)$
From (2), express y in terms of x and substitute into (4) to solve for x.
$y = \dfrac{6 - 2x}{7}$; substituting: $8x + \dfrac{5(6 - 2x)}{7} = 24 \Rightarrow 56x + 30 - 10x = 168 \Rightarrow 46x = 138 \Rightarrow x = 3$.
Back-substitute to find y and z.
$y = \dfrac{6 - 6}{7} = 0, \quad z = 5 - 3 - 0 = 2.$

Final Answer

$\vec{r} = 3\hat{i} + 2\hat{k}$

Common Mistake: Trying to solve all three equations simultaneously by inspection instead of systematically eliminating one variable at a time – this is where sign errors creep in.

Quick Recall: Convert each dot-product condition into a linear equation in x, y, z, then solve the resulting system by elimination/substitution like any ordinary system of equations.

Type 35

Proving a triangle (given by vertex coordinates or position vectors) is right-angled using the dot product: two side vectors from the same vertex are perpendicular exactly when their dot product is zero.

★ **Board Favorite** – Dot-product right-angle proofs for a triangle are a recurring long-answer ISC pattern, testing both the proof and the follow-up angle calculation.

Source: Exercise 10.2, Q30

Given: Vertices of a triangle are $A(-1,-1,2)$, $B(2,b,5)$... i.e. $A(-1,3,2)$? Use as printed: $A(2,-1,1)$, $B(1,-3,-5)$, $C(3,-4,-4)$... *(main example below uses $A(-1,3,2), B(2,3,5), C(3,5,-2)$ as printed in the book).* If the vertices of a triangle are $A(-1,3,2)$, $B(2,3,5)$, $C(3,5,-2)$, prove by vector method it is right angled at A. Also find the other two angles.

Solution:

Find the two side vectors from vertex A, since that is the vertex we suspect is the right angle.
$\vec{AB} = (3,0,3), \qquad \vec{AC} = (4,2,-4)$.
Take their dot product – if it is zero, the triangle is right-angled at A.
$\vec{AB}\cdot\vec{AC} = (3)(4)+(0)(2)+(3)(-4) = 12+0-12 = 0$.
Since $\vec{AB}\cdot\vec{AC} = 0$, $\triangle ABC$ is right-angled at A.
For the remaining angles, first find all three side lengths.
$|AB| = \sqrt{9+0+9} = 3\sqrt{2}, \quad |AC| = \sqrt{16+4+16} = 6, \quad |BC| = \sqrt{1+4+49} = 3\sqrt{6}$
(check: $|AB|^2+|AC|^2 = 18+36 = 54 = |BC|^2$ – confirms the right angle at A).
Find angle B using vectors $\vec{BA}$ and $\vec{BC}$, both starting at B.
$\vec{BA} = (-3,0,-3)$, $\vec{BC} = (1,2,-7)$; $\vec{BA}\cdot\vec{BC} = -3+0+21 = 18$.
$$\cos B = \frac{18}{(3\sqrt{2})(3\sqrt{6})} = \frac{18}{9\sqrt{12}} = \frac{2}{\sqrt{12}} = \frac{1}{\sqrt{3}}.$$
Since $A+B+C = 180^\circ$ and $A = 90^\circ$, $C = 90^\circ - B$, so $\cos C = \sin B = \sqrt{1-\frac{1}{3}} = \sqrt{\frac{2}{3}}$.

Final Answer

Right angled at A; $\quad \cos B = \dfrac{1}{\sqrt{3}}, \quad \cos C = \sqrt{\dfrac{2}{3}}$

Common Mistake: Using $\vec{AB}$ and $\vec{BC}$ (which don't share a common starting vertex) to test the angle at B instead of $\vec{BA}$ and $\vec{BC}$ – angle-at-a-vertex tests always need both vectors starting from that same vertex.

Alternate Board-Style Example

Source: Exercise 10.2, Q36(ii)

Given: Vertices of a triangle are $A(2,-1,1)$, $B(1,-3,-5)$, $C(3,-4,-4)$. Prove by vector method that it is right-angled.

Solution:

Find all three side vectors and test each pair for perpendicularity.
$\vec{BC} = (2,-1,1), \quad \vec{CA} = (-1,3,5)$.
Test the pair meeting at C: $\vec{CB} = (-2,1,-1)$ and $\vec{CA} = (-1,3,5)$.
$\vec{CB}\cdot\vec{CA} = (-2)(-1)+(1)(3)+(-1)(5) = 2+3-5 = 0$.
Since this dot product is zero, $\triangle ABC$ is right-angled at C.

Final Answer

$\triangle ABC$ is right-angled at C.

Quick Recall: To prove a right angle at a vertex, form the two side vectors starting from that vertex and check their dot product is zero.

Type 36

Resolving a vector $\vec{\beta}$ into a component $\vec{\beta}_1$ parallel to a given vector $\vec{\alpha}$, and a component $\vec{\beta}_2$ perpendicular to $\vec{\alpha}$, using the projection formula.

Formula Used: $\vec{\beta}_1 = \dfrac{\vec{\beta}\cdot\vec{\alpha}}{|\vec{\alpha}|^2}\,\vec{\alpha}$ (parallel part), $\vec{\beta}_2 = \vec{\beta} - \vec{\beta}_1$ (perpendicular part)

Source: Exercise 10.2, Q37

Given: $\vec{\alpha} = 3\hat{i} + 4\hat{j} + 5\hat{k}$, $\vec{\beta} = 2\hat{i} + \hat{j} - 4\hat{k}$. Express $\vec{\beta}$ in the form $\vec{\beta} = \vec{\beta}_1 + \vec{\beta}_2$ where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$.

Solution:

The parallel component is the vector projection of $\vec{\beta}$ onto $\vec{\alpha}$; compute the scalar factor first.

$\vec{\beta}\cdot\vec{\alpha} = (2)(3) + (1)(4) + (-4)(5) = 6 + 4 - 20 = -10, \quad |\vec{\alpha}|^2 = 9 + 16 + 25 = 50.$

Scale $\vec{\alpha}$ by this factor to get $\vec{\beta}_1$.

$\vec{\beta}_1 = \dfrac{-10}{50}\vec{\alpha} = -\dfrac{1}{5}(3\hat{i} + 4\hat{j} + 5\hat{k}) = -\dfrac{3}{5}\hat{i} - \dfrac{4}{5}\hat{j} - \hat{k}.$

Find the perpendicular component by subtracting $\vec{\beta}_1$ from $\vec{\beta}$, since the two components must add back to $\vec{\beta}$.

$\vec{\beta}_2 = \vec{\beta} - \vec{\beta}_1 = \left(2 + \dfrac{3}{5}\right)\hat{i} + \left(1 + \dfrac{4}{5}\right)\hat{j} + (-4 + 1)\hat{k} = \dfrac{13}{5}\hat{i} + \dfrac{9}{5}\hat{j} - 3\hat{k}.$

Final Answer

$\vec{\beta}_1 = -\dfrac{1}{5}(3\hat{i} + 4\hat{j} + 5\hat{k}), \qquad \vec{\beta}_2 = \dfrac{1}{5}(13\hat{i} + 9\hat{j} - 15\hat{k})$

Common Mistake: Forgetting to divide by $|\vec{\alpha}|^2$ (not $|\vec{\alpha}|$) in the projection formula – this is the single most common slip in resolving vectors into components.

Quick Recall: Parallel part $= \dfrac{\vec{\beta}\cdot\vec{\alpha}}{|\vec{\alpha}|^2}\vec{\alpha}$; perpendicular part = original vector minus the parallel part.

Type 37

Finding a vector that is perpendicular to one given vector and coplanar with two other given vectors – write it as a linear combination $\lambda\vec{a} + \mu\vec{b}$ of the two vectors it must be coplanar with, then impose perpendicularity to the third to relate λ, μ, and finally scale to any required magnitude.

Formula Used: Coplanar with $\vec{a}, \vec{b}$: $\quad \vec{r} = \lambda\vec{a} + \mu\vec{b}$

Source: Exercise 10.3, Q41

Given: Find a vector of magnitude 5 units which is coplanar with vectors $3\hat{i} - \hat{j} - \hat{k}$ and $\hat{i} + \hat{j} - 2\hat{k}$, and is perpendicular to the vector $2\hat{i} + 2\hat{j} + \hat{k}$.

Solution:

Since the required vector must be coplanar with the two given vectors, write it as their linear combination.
$\vec{r} = \lambda(3\hat{i} - \hat{j} - \hat{k}) + \mu(\hat{i} + \hat{j} - 2\hat{k}) = (3\lambda + \mu)\hat{i} + (-\lambda + \mu)\hat{j} + (-\lambda - 2\mu)\hat{k}$.
Impose perpendicularity to $2\hat{i} + 2\hat{j} + \hat{k}$ by setting the dot product to zero.
$2(3\lambda+\mu)+2(-\lambda+\mu)+1(-\lambda-2\mu) = 6\lambda+2\mu-2\lambda+2\mu-\lambda-2\mu = 3\lambda+2\mu = 0 \Rightarrow \mu = -\dfrac{3\lambda}{2}$.
Choose a convenient value of λ (say $\lambda = 2$, so $\mu = -3$) to get a vector in the right direction.
$\vec{r} = (6 - 3)\hat{i} + (-2 - 3)\hat{j} + (-2 + 6)\hat{k} = 3\hat{i} - 5\hat{j} + 4\hat{k}$.
Scale this to the required magnitude 5: find its current magnitude first.
$|\vec{r}| = \sqrt{9 + 25 + 16} = \sqrt{50} = 5\sqrt{2}$.
Multiply by $\dfrac{5}{5\sqrt{2}} = \dfrac{1}{\sqrt{2}}$ to get the final vector (keeping $\pm$ since either direction along this line works).
$\vec{r}_{\text{final}} = \pm\dfrac{1}{\sqrt{2}}(3\hat{i} - 5\hat{j} + 4\hat{k})$.

Final Answer

$\vec{r} = \pm\dfrac{1}{\sqrt{2}}(3\hat{i} - 5\hat{j} + 4\hat{k})$

Common Mistake: Forgetting the final rescaling step to match the required magnitude – the coplanar + perpendicular conditions only fix the *direction*, not the length.

Quick Recall: "Coplanar with $\vec{a}, \vec{b}$" means write $\vec{r} = \lambda\vec{a} + \mu\vec{b}$; then use the perpendicularity condition to fix the ratio $\lambda : \mu$, and finally rescale to any required magnitude.

Type 38

Finding the length of a parallelogram's diagonal using vector combinations of its sides (with a known angle between the base vectors), and testing whether two given vectors are orthogonal.

Source: Exercise 10.2, Q44, Q45

Given: (Q44) Parallelogram $ABCD$ has adjacent sides $\vec{AB} = 3\vec{a} - 5\vec{b}$ and $\vec{AD} = \vec{a} - 2\vec{b}$, with $|\vec{a}| = \sqrt{2}$, $|\vec{b}| = \sqrt{8}$, and angle between $\vec{a}, \vec{b}$ is $\frac{\pi}{3}$. Find $|\vec{BD}|$. (Q45) Determine whether $\vec{u} = \hat{i} + 3\hat{j} + \hat{k}$ and $\vec{v} = 5\hat{i} - 2\hat{j} + \hat{k}$ are orthogonal.

Solution:

Case: Diagonal length (Q44)

The diagonal $\vec{BD}$ goes from B to D, i.e. $\vec{BD} = \vec{AD} - \vec{AB}$.
$\vec{BD} = (\vec{a} - 2\vec{b}) - (3\vec{a} - 5\vec{b}) = -2\vec{a} + 3\vec{b}$.
Square this to find $|\vec{BD}|^2$, expanding with the dot product; first find $\vec{a} \cdot \vec{b}$ using the given angle.
$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos 60^\circ = \sqrt{2} \cdot \sqrt{8} \cdot \frac{1}{2} = 4 \cdot \frac{1}{2} = 2$.
$|\vec{BD}|^2 = |-2\vec{a} + 3\vec{b}|^2 = 4|\vec{a}|^2 - 12(\vec{a} \cdot \vec{b}) + 9|\vec{b}|^2 = 4(2) - 12(2) + 9(8) = 8 - 24 + 72 = 56$.
$|\vec{BD}| = \sqrt{56} = 2\sqrt{14}$.

Case: Orthogonality test (Q45)

Two vectors are orthogonal exactly when their dot product is zero – compute it directly.
$\vec{u} \cdot \vec{v} = (1)(5) + (3)(-2) + (1)(1) = 5 - 6 + 1 = 0$.
Since the dot product is zero, $\vec{u}$ and $\vec{v}$ are orthogonal.

Final Answer

$|\vec{BD}| = 2\sqrt{14}$; $\quad \vec{u}, \vec{v}$ are orthogonal (Yes)

Common Mistake: Getting the sign of the middle (cross) term wrong when squaring $-2\vec{a} + 3\vec{b}$ – it must be $2 \times (-2) \times 3 = -12$ times $\vec{a} \cdot \vec{b}$, not $+12$.

Quick Recall: Diagonal vectors in a parallelogram are sums/differences of the two adjacent side vectors – square them using the dot product to get length^2; orthogonal just means dot product $= 0$.

Type 39

Solving for an unknown parameter that preserves a vector's magnitude under a stated transformation (e.g. rotation), and solving for natural-number unknowns from a projection condition combined with a given magnitude.

Source: Exercise 10.2, Q46, Q47

Given: (Q46) $\vec{u} = (t-1)\hat{i} + 2\hat{j} + (t-3)\hat{k}$ is rotated about the origin to obtain $\vec{v} = \hat{i}+(t-1)\hat{j}+2\hat{k}$; find the possible values of t. (Q47) The projection of $\vec{a} = 3\hat{i}+q\hat{j}-\hat{k}$ on $\vec{b} = \hat{i} + \sqrt{2}\hat{j} + p\hat{k}$ is 1, where $|\vec{b}| = \sqrt{12}$; find p and q.

Solution:

Case: Rotation problem (Q46)

A rotation does not change a vector's length, so set $|\vec{u}|^2 = |\vec{v}|^2$.
$(t-1)^2 + 4 + (t-3)^2 = 1 + (t-1)^2 + 4$.
The $(t-1)^2 + 4$ terms appear on both sides and cancel, leaving a simple equation in $t-3$.
$(t-3)^2 = 1 \Rightarrow t-3 = \pm 1 \Rightarrow t = 4$ or $t = 2$.

Case: Natural-number unknowns from projection (Q47)

First use the given magnitude of $\vec{b}$ to find p.
$|\vec{b}|^2 = 1 + 2 + p^2 = 12 \Rightarrow p^2 = 9 \Rightarrow p = 3$ (taking the natural-number value).

Now compute $\vec{a}\cdot\vec{b}$ in terms of q (using $p = 3$), and use the projection condition $\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = 1$.

$\vec{a}\cdot\vec{b} = 3(1) + q(\sqrt{2}) + (-1)(3) = \sqrt{2}\,q$.
$\dfrac{\sqrt{2}\,q}{\sqrt{12}} = 1 \Rightarrow \sqrt{2}\,q = \sqrt{12} = 2\sqrt{3} \Rightarrow q = \dfrac{2\sqrt{3}}{\sqrt{2}} = \sqrt{6}$.

Final Answer

$t = 2$ or $t = 4$; $p = 3,\ q = \sqrt{6}$

Common Mistake: In the rotation problem, forgetting that rotation preserves magnitude but NOT direction – do not try to match individual components, only the overall length.

Quick Recall: "Rotation preserves magnitude" $\Rightarrow$ equate $|\vec{u}|^2 = |\vec{v}|^2$ only; for projection-based unknowns, use the given magnitude to pin down one unknown first, then substitute into the projection equation for the rest.

V. Vector (Cross) Product

Type 40

Finding the magnitude of a cross product from components, or finding the angle between two vectors when their magnitudes and $|\vec{a} \times \vec{b}|$ are given.

Formula Used: $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

Formula Used: $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$

Source: Exercise 10.3, Q1(i), Q2(i)

Given: (i) $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} - 2\hat{k}$; find $|\vec{a} \times \vec{b}|$. (ii) $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{a} \times \vec{b}| = \sqrt{3}$; find the angle between $\vec{a}$ and $\vec{b}$.

Solution:

Case: Magnitude of cross product

Set up the determinant with $\hat{i}, \hat{j}, \hat{k}$ in the first row and the two vectors' components below.

$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 3 \\ 3 & 5 & -2 \end{vmatrix} = \hat{i}(1 \cdot (-2) - 3 \cdot 5) - \hat{j}(2 \cdot (-2) - 3 \cdot 3) + \hat{k}(2 \cdot 5 - 1 \cdot 3)$.

Simplify each cofactor carefully.

$= \hat{i}(-2 - 15) - \hat{j}(-4 - 9) + \hat{k}(10 - 3) = -17\hat{i} + 13\hat{j} + 7\hat{k}$.

Take the magnitude of this resulting vector.

$|\vec{a} \times \vec{b}| = \sqrt{289 + 169 + 49} = \sqrt{507}$.

Case: Angle from $|\vec{a} \times \vec{b}|$

Substitute directly into $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$.

$\sin\theta = \dfrac{\sqrt{3}}{1 \cdot 2} = \dfrac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ$.

Final Answer

$|\vec{a} \times \vec{b}| = \sqrt{507}; \qquad \theta = 60^\circ$

Common Mistake: Sign errors in the middle ($\hat{j}$) cofactor of the determinant – remember the $\hat{j}$ term always carries an extra negative sign in the expansion.

Quick Recall: Cross product magnitude comes from the $\hat{i}, \hat{j}, \hat{k}$ determinant; for an angle from $|\vec{a} \times \vec{b}|$, use $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$ (note: sine, not cosine).

Type 41

Finding an unknown scalar that makes two given vectors parallel, using the fact that parallel vectors have a zero cross product – equivalent to their components being proportional.

Formula Used: $\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0} \iff$ components proportional

Source: Exercise 10.3, Q5(i)

Given: $(2\hat{i} + 6\hat{j} + 14\hat{k}) \times (\hat{i} - \lambda\hat{j} + 7\hat{k}) = \vec{0}$. Find λ.

Solution:

A zero cross product means the two vectors are parallel, so their corresponding components must be proportional.

$\dfrac{2}{1} = \dfrac{6}{-\lambda} = \dfrac{14}{7}$.

Both the first and third ratios equal 2, confirming consistency; use this common ratio to solve for λ.

$\dfrac{6}{-\lambda} = 2 \Rightarrow -\lambda = 3 \Rightarrow \lambda = -3$.

Final Answer

$\lambda = -3$

Common Mistake: Missing the negative sign already present in front of λ in the given vector, causing a sign flip in the final answer.

Quick Recall: Cross product $= \vec{0}$ just means "components are proportional" – set up the ratio equation exactly like the parallel-vectors dot-product-free test.

Type 42

Finding a unit vector perpendicular to two given vectors, and answering short conceptual questions about what $\vec{a} \times \vec{b} = \vec{0}$ combined with $\vec{a} \cdot \vec{b} = 0$ implies.

Formula Used: $\hat{n} = \pm\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$

Source: Exercise 10.3, Q6(i), Q8(i)

Given: (i) Find a unit vector perpendicular to $\hat{\imath}+2\hat{\jmath}-\hat{k}$ and $2\hat{\imath}+3\hat{\jmath}+\hat{k}$. (ii) What conclusion can you draw about $\vec{a},\vec{b}$ when $\vec{a}\times\vec{b}=\vec{0}$ AND $\vec{a}\cdot\vec{b}=0$?

Solution:

Case: Unit vector perpendicular to two vectors

Compute the cross product of the two given vectors, since $\vec{a}\times\vec{b}$ is always perpendicular to both.

$\vec{a}\times\vec{b} = \begin{vmatrix}\hat{\imath} & \hat{\jmath} & \hat{k}\\ 1 & 2 & -1\\ 2 & 3 & 1\end{vmatrix} = \hat{\imath}(2+3)-\hat{\jmath}(1+2)+\hat{k}(3-4) = 5\hat{\imath}-3\hat{\jmath}-\hat{k}.$

Find its magnitude and divide to normalise.

$|\vec{a}\times\vec{b}| = \sqrt{25+9+1}=\sqrt{35}$, so $\hat{n}=\pm\dfrac{5\hat{\imath}-3\hat{\jmath}-\hat{k}}{\sqrt{35}}$.

Case: Conceptual conclusion

$\vec{a}\times\vec{b}=\vec{0}$ alone would mean $\vec{a}\parallel\vec{b}$ (or one is zero); but $\vec{a}\cdot\vec{b}=0$ alone would mean $\vec{a}\perp\vec{b}$ (or one is zero). A vector cannot be simultaneously parallel and perpendicular to a non-zero vector, so both conditions holding together force one of $\vec{a},\vec{b}$ to be the zero vector.
So the conclusion is: $\vec{a}=\vec{0}$ or $\vec{b}=\vec{0}$.

Final Answer

(i) $\hat{n}=\pm\dfrac{1}{\sqrt{35}}(5\hat{\imath}-3\hat{\jmath}-\hat{k})$ (ii) $\vec{a}=\vec{0}$ or $\vec{b}=\vec{0}$

Common Mistake: Forgetting the $\pm$ sign on a "unit vector perpendicular to two vectors" question – both the cross product and its negative are valid perpendicular directions.

Quick Recall: Cross product of two vectors gives a vector perpendicular to both – normalise it for a unit vector; both $\times = 0$ and $\cdot = 0$ together force one of the vectors to be zero.

Type 43

Evaluating standard combined dot/cross-product expressions built from the unit vectors $\hat{\imath},\hat{\jmath},\hat{k}$ – recall the basic cross-product table ($\hat{\imath}\times\hat{\jmath}=\hat{k}$, etc.) directly rather than using determinants.

Formula Used: $\hat{i}\times\hat{j}=\hat{k},\quad \hat{j}\times\hat{k}=\hat{i},\quad \hat{k}\times\hat{i}=\hat{j},\qquad \hat{i}\cdot\hat{j}=\hat{j}\cdot\hat{k}=\hat{k}\cdot\hat{i}=0$

Source: Exercise 10.3, Q10(i), Q10(ii)

Given: Find (i) $(\hat{i}\times\hat{j})\cdot\hat{k}+\hat{i}\cdot\hat{j}$ (ii) $(\hat{j}\times\hat{k})\cdot\hat{i}+(\hat{i}\times\hat{k})\cdot\hat{j}+(\hat{i}\times\hat{j})\cdot\hat{k}$.

Solution:

Recall the standard cyclic cross-product identities and that dot products of distinct axis unit vectors vanish.
(i) $\hat{i}\times\hat{j}=\hat{k}$, so $(\hat{i}\times\hat{j})\cdot\hat{k}=\hat{k}\cdot\hat{k}=1$; and $\hat{i}\cdot\hat{j}=0$. Sum $=1+0=1$.
(ii) $\hat{j}\times\hat{k}=\hat{i}\Rightarrow(\hat{j}\times\hat{k})\cdot\hat{i}=\hat{i}\cdot\hat{i}=1$.
$\hat{i}\times\hat{k}=-\hat{j}$ (reverse of the cyclic order $\hat{k}\times\hat{i}=\hat{j}$) $\Rightarrow(\hat{i}\times\hat{k})\cdot\hat{j}=-\hat{j}\cdot\hat{j}=-1$.
$\hat{i}\times\hat{j}=\hat{k}\Rightarrow(\hat{i}\times\hat{j})\cdot\hat{k}=\hat{k}\cdot\hat{k}=1$.
Sum $=1-1+1=1$.

Final Answer

(i) 1 (ii) 1

Common Mistake: Getting the sign wrong on non-cyclic products like $\hat{i}\times\hat{k}$ – it equals $-\hat{j}$, not $+\hat{j}$, since the cyclic order is $\hat{i}\to\hat{j}\to\hat{k}\to\hat{i}$.

Quick Recall: Memorise the cyclic triple $\hat{i}\times\hat{j}=\hat{k},\ \hat{j}\times\hat{k}=\hat{i},\ \hat{k}\times\hat{i}=\hat{j}$ – reversing any pair flips the sign.

Type 44

Finding the area of a triangle or parallelogram from vectors – using two adjacent sides directly, using vertex coordinates (build the side vectors first), or using the two diagonals.

★ **Board Favorite –** Area-by-cross-product is one of the most frequently tested ISC patterns in this chapter, in all three forms: sides, vertices, and diagonals (seen in ISC 2020, 2023, 2025).

Formula Used: Area of parallelogram $=|\vec{a}\times\vec{b}|$ (adjacent sides)

Formula Used: Area of triangle $=\frac{1}{2}|\vec{a}\times\vec{b}|$ (two sides from a vertex)

Formula Used: Area of parallelogram $=\frac{1}{2}|\vec{d_1}\times\vec{d_2}|$ (diagonals)

Source: Exercise 10.3, Q11(i)

Given: Find the area of the parallelogram whose adjacent sides are $3\hat{i}+\hat{j}+4\hat{k}$ and $\hat{i}-\hat{j}+\hat{k}$.

Solution:

Compute the cross product of the two adjacent side vectors, since its magnitude directly gives the parallelogram's area.

$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 1 & 4\\ 1 & -1 & 1\end{vmatrix}=\hat{i}(1+4)-\hat{j}(3-4)+\hat{k}(-3-1)=5\hat{i}+\hat{j}-4\hat{k}.$

Take the magnitude of this vector – that is the area.

$|\vec{a}\times\vec{b}|=\sqrt{25+1+16}=\sqrt{42}.$

Final Answer

Area $=\sqrt{42}$ sq. units

Common Mistake: Forgetting to halve the cross product's magnitude when the question asks for a *triangle*'s area instead of a parallelogram's – always check which shape is being asked for.

Alternate Board-Style Example

Source: Exercise 10.3, Q23 (ISC 2020)

Given: Using vectors, find the area of the triangle whose vertices are $A(3,-1,2)$, $B(1,-1,-3)$, $C(4,-3,1)$.

Solution:

Build two side vectors from a common vertex A.

$\vec{AB}=(-2,0,-5), \qquad \vec{AC}=(1,-2,-1).$

Compute their cross product.

$\vec{AB}\times\vec{AC}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -2 & 0 & -5\\ 1 & -2 & -1\end{vmatrix}=\hat{i}(0-10)-\hat{j}(2+5)+\hat{k}(4-0)=-10\hat{i}-7\hat{j}+4\hat{k}.$

Find its magnitude, then halve it for the triangle's area.

$|\vec{AB}\times\vec{AC}|=\sqrt{100+49+16}=\sqrt{165}.$

Area $=\dfrac{1}{2}\sqrt{165}.$

Final Answer

Area $=\dfrac{1}{2}\sqrt{165}$ sq. units

Quick Recall: Cross product magnitude = parallelogram area; half of it = triangle area; from vertices, build two side vectors from one common vertex first.

Type 45

Evaluating magnitude/combination expressions built from cross products, such as $|\vec{b} \times 2\vec{a}|$ – simplify using the scalar-multiple property of the cross product before computing.

Formula Used: $\vec{b} \times (k\vec{a}) = k(\vec{b} \times \vec{a})$

Source: Exercise 10.3, Q13(i)

Given: $\vec{a} = 4\hat{i} + 3\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} + 2\hat{k}$. Find $|\vec{b} \times 2\vec{a}|$.

Solution:

Pull the scalar 2 out of the cross product first, since scalars factor straight out of either side.
$|\vec{b} \times 2\vec{a}| = 2|\vec{b} \times \vec{a}|$.
Compute $\vec{b} \times \vec{a}$ using the determinant.

$$\vec{b} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & 2 \\ 4 & 3 & 2 \end{vmatrix} = \hat{i}(0-6) - \hat{j}(6-8) + \hat{k}(9-0) = -6\hat{i} + 2\hat{j} + 9\hat{k}.$$

Find its magnitude, then double it.
$|\vec{b} \times \vec{a}| = \sqrt{36+4+81} = \sqrt{121} = 11$, so $|\vec{b} \times 2\vec{a}| = 2 \times 11 = 22$.

Final Answer

22

Common Mistake: Computing $\vec{a} \times \vec{b}$ instead of $\vec{b} \times \vec{a}$ – although the magnitude is the same either way (since $|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{a}|$), always match the order asked for if the direction/sign matters elsewhere in the question.

Quick Recall: Pull out scalar multiples from a cross product before computing – $|\vec{b} \times k\vec{a}| = |k| \cdot |\vec{b} \times \vec{a}|$.

Type 46

Finding $\sin\theta$ between two vectors directly from their cross product, without first finding θ itself.

Formula Used: $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$

Source: Exercise 10.3, Q15

Given: If θ is the angle between $\hat{\imath} - 2\hat{\jmath} + 3\hat{k}$ and $3\hat{\imath} - 2\hat{\jmath} + \hat{k}$, find $\sin\theta$.

Solution:

Compute the cross product of the two vectors.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & -2 & 3 \\ 3 & -2 & 1 \end{vmatrix} = \hat{\imath}(-2+6) - \hat{\jmath}(1-9) + \hat{k}(-2+6) = 4\hat{\imath} + 8\hat{\jmath} + 4\hat{k}.$$

Find its magnitude, and also the magnitudes of $\vec{a}$ and $\vec{b}$.

$|\vec{a} \times \vec{b}| = \sqrt{16+64+16} = \sqrt{96} = 4\sqrt{6}$; $|\vec{a}| = \sqrt{1+4+9} = \sqrt{14}$; $|\vec{b}| = \sqrt{9+4+1} = \sqrt{14}$.

Substitute into the sine formula.

$$\sin\theta = \frac{4\sqrt{6}}{\sqrt{14}\cdot\sqrt{14}} = \frac{4\sqrt{6}}{14} = \frac{2\sqrt{6}}{7}.$$

Final Answer

$$\sin\theta = \frac{2\sqrt{6}}{7}$$

Common Mistake: Using the dot-product (cosine) formula by mistake when the question specifically asks for $\sin\theta$ – sine always comes from the cross product's magnitude.

Quick Recall: $\sin\theta$ between two vectors $= \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$ – no need to find θ itself first.

Type 47

Finding a vector of a given magnitude that is perpendicular to two given vectors – the plain version, and a variant with an extra dot-product condition that fixes the sign/scale directly.

Source: Exercise 10.3, Q17(i)

Given: Find a vector of magnitude 6 units perpendicular to both $2\hat{\imath} - \hat{\jmath} + 2\hat{k}$ and $4\hat{\imath} - \hat{\jmath} + 3\hat{k}$. (Exemplar)

Solution:

The cross product of the two vectors is automatically perpendicular to both, so compute it first.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -1 & 2 \\ 4 & -1 & 3 \end{vmatrix} = \hat{\imath}(-3+2) - \hat{\jmath}(6-8) + \hat{k}(-2+4) = -\hat{\imath} + 2\hat{\jmath} + 2\hat{k}.$$

Find its magnitude, then figure out the scale factor needed to reach magnitude 6.

$|\vec{a} \times \vec{b}| = \sqrt{1+4+4} = 3$.

Scale factor $= \dfrac{6}{3} = 2$; multiply the cross product by this factor (keeping $\pm$, since perpen-

dicular vectors exist in both senses).
$\vec{r} = \pm 2(-\hat{i} + 2\hat{j} + 2\hat{k}) = \pm(-2\hat{i} + 4\hat{j} + 4\hat{k})$.

Final Answer

$\vec{r} = \pm 2(\hat{i} - 2\hat{j} - 2\hat{k})$

Common Mistake: Stopping at the cross product itself without rescaling to the magnitude actually asked for in the question.

Quick Recall: Perpendicular to two vectors $\Rightarrow$ use their cross product; then rescale it to whatever magnitude is required.

Type 48

Proving the identity $|\vec{a} \times \vec{b}| = (\vec{a} \cdot \vec{b}) \tan\theta$, connecting the two product definitions through the angle θ between the vectors.

Source: Exercise 10.3, Q21

Given: Define $\vec{a} \times \vec{b}$ and prove that $|\vec{a} \times \vec{b}| = (\vec{a} \cdot \vec{b}) \tan\theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$.

Solution:

Write down the two defining formulas for magnitude of the cross product and the dot product, both in terms of θ.
$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta, \qquad \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta.$
Divide the first equation by the second, since the common factor $|\vec{a}||\vec{b}|$ cancels neatly.
$$\frac{|\vec{a} \times \vec{b}|}{\vec{a} \cdot \vec{b}} = \frac{|\vec{a}||\vec{b}| \sin\theta}{|\vec{a}||\vec{b}| \cos\theta} = \tan\theta.$$
Rearranging gives exactly the required identity.
$|\vec{a} \times \vec{b}| = (\vec{a} \cdot \vec{b}) \tan\theta.$

Final Answer

Proved: $|\vec{a} \times \vec{b}| = (\vec{a} \cdot \vec{b}) \tan\theta$

Common Mistake: Trying to prove this using components for a specific example instead of the general definitions – this identity must be shown to hold for ANY $\vec{a}, \vec{b}$, so only the general $|\vec{a}||\vec{b}| \sin\theta$ and $|\vec{a}||\vec{b}| \cos\theta$ forms will do.

Quick Recall: This identity is just "divide the cross-product-magnitude formula by the dot-product formula" – the common factor $|\vec{a}||\vec{b}|$ cancels to leave $\tan\theta$.

Type 49

Proving/deriving conditions connecting equal vectors to cross products – e.g. showing $\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$ under $\vec{b} = \vec{c}$, or the converse-style proof that $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ forces $\vec{b} = \vec{c}$ or $\vec{a} \parallel (\vec{b} - \vec{c})$.

Source: Exercise 10.3, Q31, Q26 (ISC 2024)

Given: (Q31) Given $\vec{b} = \vec{c}$, derive the condition under which $\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$. (Q26) If $\vec{a}, \vec{b}, \vec{c}$ are non-zero vectors such that $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$, prove that either $\vec{b} = \vec{c}$ or $\vec{a}$ and $(\vec{b} - \vec{c})$ are parallel.

Solution:

Case: Deriving the condition (Q31)

Since $\vec{b} = \vec{c}$, replace $\vec{c}$ by $\vec{b}$ in the target equation and simplify.
$\vec{a} \times \vec{b} = \vec{c} \times \vec{a} = \vec{b} \times \vec{a} = -(\vec{a} \times \vec{b})$.
This gives $\vec{a} \times \vec{b} = -(\vec{a} \times \vec{b}) \Rightarrow 2(\vec{a} \times \vec{b}) = \vec{0} \Rightarrow \vec{a} \times \vec{b} = \vec{0}$.
So the condition is that $\vec{a}$ and $\vec{b}$ must themselves be parallel (or one is zero).

Case: Proving the converse-style result (Q26)

Bring both cross products to one side and factor using distributivity of the cross product.
$\vec{a} \times \vec{b} - \vec{a} \times \vec{c} = \vec{0} \Rightarrow \vec{a} \times (\vec{b} - \vec{c}) = \vec{0}$.
A zero cross product means the two vectors involved are parallel (or one is the zero vector); since $\vec{a}$ is given non-zero, either $(\vec{b} - \vec{c}) = \vec{0}$ (i.e. $\vec{b} = \vec{c}$) or $\vec{a} \parallel (\vec{b} - \vec{c})$.

Final Answer

(Q31) Condition: $\vec{a} \times \vec{b} = \vec{0}$ (Q26) Proved: $\vec{b} = \vec{c}$ or $\vec{a} \parallel (\vec{b} - \vec{c})$

Common Mistake: Forgetting that $\vec{c} \times \vec{a} = -(\vec{a} \times \vec{c})$ when swapping the order of a cross product – cross product is anti-commutative, unlike the dot product.

Quick Recall: Cross product is anti-commutative ($\vec{x} \times \vec{y} = -\vec{y} \times \vec{x}$) and distributes over subtraction – both proof types here are built entirely from those two facts.

Type 50

Resolving a vector $\vec{\beta}$ into a part $\vec{\beta}_1$ parallel to a given vector $\vec{\alpha}$ and a part $\vec{\beta}_2$ perpendicular to $\vec{\alpha}$ (cross-product-flavoured version), then computing $\vec{\beta}_1 \times \vec{\beta}_2$.

Source: Exercise 10.3, Q32

Given: $\vec{\alpha} = 3\hat{i} + \hat{j}$, $\vec{\beta} = 2\hat{i} - \hat{j} + 3\hat{k}$. Given $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$ where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$: (a) find $\vec{\beta}_1$ (b) find $\vec{\beta}_2$ (c) hence find $\vec{\beta}_1 \times \vec{\beta}_2$.

Solution:

Case: Part (a) – parallel component

Use the projection formula to find $\vec{\beta}_1$, the part of $\vec{\beta}$ along $\vec{\alpha}$.

$\vec{\beta} \cdot \vec{\alpha} = (2)(3) + (-1)(1) + (3)(0) = 6 - 1 = 5, \quad |\vec{\alpha}|^2 = 9 + 1 = 10.$

$\vec{\beta}_1 = \dfrac{5}{10}\vec{\alpha} = \dfrac{1}{2}(3\hat{i} + \hat{j}) = \dfrac{3}{2}\hat{i} + \dfrac{1}{2}\hat{j}.$

Case: Part (b) – perpendicular component

Since the question defines $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$ (note the minus sign, not the usual plus), rearrange to solve for $\vec{\beta}_2$.

$\vec{\beta}_2 = \vec{\beta}_1 - \vec{\beta} = \left(\dfrac{3}{2}\hat{i} + \dfrac{1}{2}\hat{j}\right) - (2\hat{i} - \hat{j} + 3\hat{k}) = -\dfrac{1}{2}\hat{i} + \dfrac{3}{2}\hat{j} - 3\hat{k}.$

Case: Part (c) – cross product $\vec{\beta}_1 \times \vec{\beta}_2$

Set up the determinant using the two components just found.

$\vec{\beta}_1 \times \vec{\beta}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{3}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{3}{2} & -3 \end{vmatrix} = \hat{i}\left(-\dfrac{3}{2} - 0\right) - \hat{j}\left(-\dfrac{9}{2} - 0\right) + \hat{k}\left(\dfrac{9}{4} + \dfrac{1}{4}\right).$

Simplify each component.

$= -\dfrac{3}{2}\hat{i} + \dfrac{9}{2}\hat{j} + \dfrac{10}{4}\hat{k} = -\dfrac{3}{2}\hat{i} + \dfrac{9}{2}\hat{j} + \dfrac{5}{2}\hat{k}.$

Final Answer

$\vec{\beta}_1 = \dfrac{3}{2}\hat{i} + \dfrac{1}{2}\hat{j}, \quad \vec{\beta}_2 = -\dfrac{1}{2}\hat{i} + \dfrac{3}{2}\hat{j} - 3\hat{k}, \quad \vec{\beta}_1 \times \vec{\beta}_2 = -\dfrac{3}{2}\hat{i} + \dfrac{9}{2}\hat{j} + \dfrac{5}{2}\hat{k}$

Common Mistake: Using the usual $\vec{\beta}_2 = \vec{\beta} - \vec{\beta}_1$ relation out of habit – always check exactly how the question defines the split ($\vec{\beta}_1 + \vec{\beta}_2$ vs $\vec{\beta}_1 - \vec{\beta}_2$) before subtracting.

Quick Recall: Find the parallel part by projection as usual, but always re-check the question's exact relation (plus or minus) before solving for the second part.

Type 51

A composite question combining several methods in one: finding side vectors $\vec{AB}, \vec{BC}$ from given position vectors, then the projection of one side on another, then the triangle's area – all from the same data set.

★ **Board Favorite –** Composite, multi-part questions combining vector subtraction, projection, and area are a favourite way for ISC to test several skills in a single high-mark question (ISC 2025).

Source: Exercise 10.3, Q33

Given: Position vectors: $\vec{OA} = 2\hat{i} - 2\hat{j} + \hat{k}$, $\vec{OB} = \hat{i} + 2\hat{j} - 2\hat{k}$, $\vec{OC} = 2\hat{i} - \hat{j} + 4\hat{k}$. (i) Calculate $\vec{AB}$ and $\vec{BC}$. (ii) Find the projection of $\vec{AB}$ on $\vec{BC}$. (iii) Find the area of $\triangle ABC$ whose sides are $\vec{AB}$ and $\vec{BC}$.

Solution:

Case: Part (i) – side vectors

Apply terminal-minus-initial to get each side vector from the given position vectors.
$\vec{AB} = \vec{OB} - \vec{OA} = (\hat{i} + 2\hat{j} - 2\hat{k}) - (2\hat{i} - 2\hat{j} + \hat{k}) = -\hat{i} + 4\hat{j} - 3\hat{k}$.
$\vec{BC} = \vec{OC} - \vec{OB} = (2\hat{i} - \hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} - 2\hat{k}) = \hat{i} - 3\hat{j} + 6\hat{k}$.

Case: Part (ii) – projection

Use the projection formula with the two vectors just found.
$\vec{AB} \cdot \vec{BC} = (-1)(1) + (4)(-3) + (-3)(6) = -1 - 12 - 18 = -31$.
$|\vec{BC}| = \sqrt{1 + 9 + 36} = \sqrt{46}$.
Projection of $\vec{AB}$ on $\vec{BC} = \dfrac{-31}{\sqrt{46}}$.

Case: Part (iii) – area of the triangle

Compute the cross product of the same two side vectors.

$$\vec{AB} \times \vec{BC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 4 & -3 \\ 1 & -3 & 6 \end{vmatrix} = \hat{i}(24 - 9) - \hat{j}(-6 + 3) + \hat{k}(3 - 4) = 15\hat{i} + 3\hat{j} - \hat{k}.$$

Find its magnitude and halve it for the triangle's area.
$|\vec{AB} \times \vec{BC}| = \sqrt{225 + 9 + 1} = \sqrt{235}$; Area $= \dfrac{1}{2}\sqrt{235}$.

Final Answer

(i) $\vec{AB} = -\hat{i} + 4\hat{j} - 3\hat{k}$, $\vec{BC} = \hat{i} - 3\hat{j} + 6\hat{k}$ (ii) Projection $= \dfrac{-31}{\sqrt{46}}$ (iii) Area $= \dfrac{1}{2}\sqrt{235}$ sq. units

Common Mistake: Re-deriving $\vec{AB}$ or $\vec{BC}$ differently for each part instead of reusing the same two vectors found in part (i) – once found, the same two vectors drive every remaining part.

Quick Recall: In a composite question, find the core vectors once (usually from position vectors), then reuse them directly for every projection/area/angle sub-part that follows.